

SYMMETRICALLY THERMALIZING UNITARIES EXIST IN EVERY DIMENSION

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ABSTRACT. Two identical quantum systems with Hamiltonian H , both in the thermal state τ_β , can be correlated by a global unitary. The energy needed to create a given amount of mutual information is bounded from below, and the bound is attained when there is a *symmetrically thermalizing unitary* (STU): a unitary U on $\mathbb{C}^d \otimes \mathbb{C}^d$ such that both marginals of $U(\tau_\beta \otimes \tau_\beta)U^\dagger$ are thermal at one common higher temperature, $\tau_{\beta'}$ with $\beta' \leq \beta$. Bakhshinezhad, Clivaz, Vitagliano, Erker, Rezakhani, Huber and Friis (2019) proved that STUs exist in local dimensions $d \leq 4$ and conjectured that they exist in every dimension. We prove this conjecture: for every d , every Hamiltonian H , every $\beta > 0$ and every $\beta' \in [0, \beta]$ a symmetrically thermalizing unitary exists. In an eigenbasis of H it can be chosen real orthogonal, and it is given by an explicit construction. The proof is elementary and does not use Kirwan’s convexity theorem. It reduces the problem to majorization conditions on the diagonals of a $d \times d$ array, writes the target as a convex combination of clipped thermal vectors, and builds, for every interval of hook lengths and every box size, an explicit symmetric “interval uniformisation” matrix from renewal squares and at most one correcting dipole; the dipole amplitude is fixed by identities for alternating harmonic sums. The complete proof, for every Hermitian H (via the case $H = \text{diag}(E)$ with sorted energies and a local unitary), has been formalised in Lean 4 with Mathlib and uses only Lean’s standard axioms. We also show that the stronger statement “every spectrum majorized by the initial one is symmetrically reachable” is false in every dimension $d \geq 4$.

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1. INTRODUCTION

1.1. Correlations at minimal energy cost. Correlations between physical systems are a resource, and creating them costs energy [11, 4, 20]. A basic instance of this trade-off is the following [2]. Two identical quantum systems A and B , each with Hamiltonian H on $\mathbb{C}^d$, are initially uncorrelated and in equilibrium with a bath at inverse temperature $\beta > 0$, so that the global state is $\tau_\beta \otimes \tau_\beta$ with

$$\tau_\beta = \tau_\beta(H) := \frac{e^{-\beta H}}{\text{Tr } e^{-\beta H}}.$$

This initial state is completely passive [17]: no energy can be extracted from it by unitaries, even from many copies. With full control of the closed composite system one may apply any global unitary U , producing $\rho' = U(\tau_\beta \otimes \tau_\beta)U^\dagger$. The invested energy $\Delta E = \text{Tr}[(H \otimes 1 + 1 \otimes H)(\rho' - \tau_\beta \otimes \tau_\beta)]$ depends only on the marginals $\rho'_A = \text{Tr}_B \rho'$ and $\rho'_B = \text{Tr}_A \rho'$. Measuring correlations by the mutual information $I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$, and using that the global von Neumann entropy is invariant under unitaries, the created correlations are $\Delta I = \Delta S_A + \Delta S_B$. Since thermal states maximise the entropy at fixed energy, and the entropy of thermal states is a concave function of their energy, an invested energy $\Delta E = 2 \text{Tr}[H(\tau_{\beta'} - \tau_\beta)]$ with $0 \leq \beta' \leq \beta$ can create at most $\Delta I \leq 2[S(\tau_{\beta'}) - S(\tau_\beta)]$, with equality if both final marginals equal $\tau_{\beta'}$ [2, Sec. II and App. A.I]. The resulting lower bound on the energy cost of correlations is therefore attained whenever the following unitaries exist.

Definition 1.1. Let H be a Hamiltonian on $\mathbb{C}^d$ and $0 \leq \beta' \leq \beta$. A unitary U on $\mathbb{C}^d \otimes \mathbb{C}^d$ is *symmetrically thermalizing* (an STU) for (H, β, β') if

$$\text{Tr}_B[U(\tau_\beta \otimes \tau_\beta)U^\dagger] = \text{Tr}_A[U(\tau_\beta \otimes \tau_\beta)U^\dagger] = \tau_{\beta'}.$$

Bakhshinezhad, Clivaz, Vitagliano, Erker, Rezakhani, Huber and Friis asked whether STUs exist for every Hamiltonian, every β and every $\beta' \leq \beta$ [2, Question 2], and conjectured that they do [2, Sec. IV]. The symmetry of the problem is essential: when the two systems have different Hamiltonians the analogous statement fails [2, Sec. II.1.1].

1.2. Previous status. STUs were known to exist when the spectrum of H is equally spaced [11] (in particular for qubits, $d = 2$), and for every Hamiltonian in local dimensions $d = 3$ and $d = 4$ [2]. For $d \geq 5$, Ref. [2] gave sufficient conditions and a framework of locally classical subspaces; the general case was still open in the PhD thesis [7]. See also [12, 16] for related questions about correlations in unitary orbits.

1.3. Main result. We prove the conjecture.

Theorem (informal version of Theorems 1 and 2). *For every local dimension d , every Hamiltonian H on $\mathbb{C}^d$, every $\beta > 0$ and every $\beta' \in [0, \beta]$ there is a symmetrically thermalizing unitary. In an eigenbasis of H it can be chosen to be a real orthogonal matrix, and it is given by an explicit construction.*

Three features of the proof deserve mention.

- *Elementary and explicit.* The only tools are majorization, the existence direction of the real Schur–Horn theorem (in its constructive form [5]), and elementary estimates for alternating series. Every object in the construction is explicit.
- *Kirwan-free.* For a fixed global spectrum, the set of pairs of marginal spectra in the unitary orbit is a convex polytope by Kirwan’s convexity theorem [13]; see [14, 8, 6] for this quantum marginal problem. Convexity reduces the conjecture to the reachability of finitely many explicit points, which gives an alternative route that we recall in Section 11.2. Our proof does not use it: we work inside one fixed combinatorial structure in which convexity is automatic.
- *Machine-checked.* The complete proof of Theorem 1 has been formalised in Lean 4 [9] with Mathlib [19]: Theorem 2, the statement for $H = \text{diag}(E)$ with sorted energies, and the reduction of an arbitrary Hermitian H to it, with $\tau_\beta(H) = e^{-\beta H} / \text{Tr } e^{-\beta H}$ defined by the matrix exponential. This includes all analytic estimates. The formalisation uses only Lean’s three standard axioms and contains no unproved statements (Section 10).

1.4. Strategy. Let $p_0 \geq p_1 \geq \dots \geq p_{d-1} > 0$ be the eigenvalues of τ_β and q those of $\tau_{\beta'}$, in the same eigenbasis of H .

- (1) *Classical reduction* (Section 3). Arrange the product eigenvectors $|i\rangle|j\rangle$ of $\tau_\beta \otimes \tau_\beta$ in a $d \times d$ array. Each of its $2d - 1$ diagonals meets every row and every column at most once, so a Schur–Horn rotation inside a diagonal keeps both marginals diagonal. Hence it suffices to find a real symmetric $d \times d$ matrix F whose diagonals are majorized by the corresponding diagonals of $p \otimes p$ and whose row sums are q . The set $\text{Reach}(p)$ of such row-sum vectors is convex.
- (2) *Clipped points* (Lemma 3.7). The vector q is a convex combination of the clipped thermal vectors $w_k = \min(p, p_k) / Z_k$, $0 \leq k \leq d - 1$. This holds for every concave nondecreasing reweighting of p , in particular for $q \propto p^{\beta' / \beta}$.
- (3) *Young diagrams* (Sections 3 and 4). With $B = \min(p, p_k)$, the difference $p \otimes p - B \otimes B$ is a nonnegative combination of symmetric Young diagrams (squares and crosses). All pieces are nonincreasing along every diagonal, so majorizations can be superposed. Each piece is replaced by an *interval uniformisation*: a symmetric matrix with entries in $[0, 1]$, the same diagonal sums as the piece, and constant row sums. The pieces are distributed over square boxes according to an explicit capacity (Hall-type) identity. This gives $w_k \in \text{Reach}(p)$.

- (4) *Interval uniformisations* (Sections 4–9). We construct them for every box size n and every interval $[\ell, h]$ of hook lengths: renewal squares, their complements, and a Toeplitz-plus-Hankel matrix (the $T/H/G$ matrix) corrected by at most one *dipole*. Diagonal sums and row sums hold by exact identities. Nonnegativity and the bound 1 reduce to one-dimensional inequalities, which follow from six facts F1–F6 about the numbers

$$a(K) = 4K^2(-1)^K \left(\ln 2 - \sum_{i=1}^K \frac{(-1)^{i-1}}{i} \right) + 1 - 2K,$$

proved with a two-sided bound on the tails of the alternating harmonic series.

Finally, the “strong form” of the problem, in which every spectrum majorized by p is symmetrically reachable, holds for $d = 3$ [2, Thm. 1] but fails for every $d \geq 4$ (Proposition 11.1). So the thermal form of the target must be used, as it is in step (2).

1.5. Organisation. Section 2 states the results precisely, including the exact Lean statement, and reduces a general Hamiltonian to the diagonal sorted case. Section 3 contains the classical reduction, Lemma T and the decomposition of $p \otimes p$. Section 4 defines interval uniformisations, proves that they suffice, and treats renewal squares and complements. Sections 5–9 construct the remaining interval uniformisations. Section 10 describes the formal verification and Section 11 contains further remarks: the counterexample to the strong form, the Kirwan route, and an outlook. Appendix A lists the polynomial identities behind F1–F4, and Appendix B lists the independent computer checks of the formulas in this paper.

2. STATEMENT OF THE RESULTS

We write $[d] = \{0, 1, \dots, d-1\}$ and use the standard basis $|i\rangle$, $i \in [d]$, of $\mathbb{C}^d$. Operators on $\mathbb{C}^d \otimes \mathbb{C}^d$ are matrices indexed by $[d] \times [d]$, and the partial traces are

$$(\mathrm{Tr}_B X)_{a,a'} = \sum_{b \in [d]} X_{(a,b),(a',b)}, \quad (\mathrm{Tr}_A X)_{b,b'} = \sum_{a \in [d]} X_{(a,b),(a,b')}.$$

For $E \in \mathbb{R}^d$ and $\beta \in \mathbb{R}$ the *Gibbs vector* is $\gamma_\beta(E)_i = e^{-\beta E_i} / \sum_j e^{-\beta E_j}$, so that $\tau_\beta(\mathrm{diag} E) = \mathrm{diag} \gamma_\beta(E)$.

Theorem 1 (STUs exist in every dimension). *Let $d \geq 1$, let H be a Hermitian operator on $\mathbb{C}^d$, and let $\beta > 0$ and $0 \leq \beta' \leq \beta$. There is a unitary U on $\mathbb{C}^d \otimes \mathbb{C}^d$ such that*

$$\mathrm{Tr}_B [U(\tau_\beta(H) \otimes \tau_\beta(H))U^\dagger] = \mathrm{Tr}_A [U(\tau_\beta(H) \otimes \tau_\beta(H))U^\dagger] = \tau_{\beta'}(H).$$

If $H = V \mathrm{diag}(E) V^\dagger$ with V unitary and $E_0 \leq \dots \leq E_{d-1}$, one can take $U = (V \otimes V) O (V \otimes V)^\dagger$ with O a real orthogonal matrix.

Theorem 2 (diagonal form). *Let $d \in \mathbb{N}$, let $E \in \mathbb{R}^d$ with $E_0 \leq E_1 \leq \dots \leq E_{d-1}$, let $\beta > 0$ and $0 \leq \beta' \leq \beta$, and put $D_\beta = \mathrm{diag} \gamma_\beta(E)$. There is a unitary U on $\mathbb{C}^d \otimes \mathbb{C}^d$, which can be taken to be a real orthogonal matrix, such that*

$$\mathrm{Tr}_B [U(D_\beta \otimes D_\beta)U^\dagger] = \mathrm{Tr}_A [U(D_\beta \otimes D_\beta)U^\dagger] = D_{\beta'}.$$

The cases $\beta' = \beta$ ($U = 1$) and $d \leq 1$ are trivial, and for $\beta = 0$ necessarily $\beta' = 0$ and $U = 1$ works. Degenerate energies are allowed.

Remark 2.1 (the formal statement). Except for the clause “which can be taken to be a real orthogonal matrix”, Theorem 2 is literally the Lean theorem `STUProof.stu_exists_unconditional` in the file `lean/STUProof/KirwanFreeFinal.lean` of the repository [1]:

```

theorem stu_exists_unconditional :
  ∀ (d : ℕ) (E : Fin d → ℝ), Monotone E →
  ∀ β β' : ℝ, 0 < β → 0 ≤ β' → β' ≤ β →
  ∃ U ∈ unitaryGroup (Fin d × Fin d) ℂ,
    partialTraceB (U * (gibbsState E β ⊗k gibbsState E β) * star U)
      = gibbsState E β' ∧
    partialTraceA (U * (gibbsState E β ⊗k gibbsState E β) * star U)
      = gibbsState E β'

```

Here `Monotone E` means $E_0 \leq \dots \leq E_{d-1}$, `star U` is $U^\dagger$, $\otimes_k$ is the Kronecker product, and the four project definitions entering the statement are

```

def gibbs (E : Fin d → ℝ) (β : ℝ) : Fin d → ℝ :=
  fun i => Real.exp (-β * E i) / ∑ j, Real.exp (-β * E j)
def gibbsState (E : Fin d → ℝ) (β : ℝ) : Matrix (Fin d) (Fin d) ℂ :=
  diagonal fun i => (gibbs E β i : ℂ)
def partialTraceB (ρ : Matrix (m × n) (m × n) ℝ) : Matrix m m ℝ :=
  Matrix.of fun a a' => ∑ b, ρ (a, b) (a', b)
def partialTraceA (ρ : Matrix (m × n) (m × n) ℝ) : Matrix n n ℝ :=
  Matrix.of fun b b' => ∑ a, ρ (a, b) (a, b')

```

The proof constructs U as a real orthogonal matrix, embedded in the complex matrices.

Proof of Theorem 1 from Theorem 2. By the spectral theorem, $H = V \text{diag}(E) V^\dagger$ with V unitary and $E_0 \leq \dots \leq E_{d-1}$ (list the eigenvalues in nondecreasing order). Then $e^{-\beta H} = V \text{diag}(e^{-\beta E_i}) V^\dagger$, hence $\tau_\beta(H) = V D_\beta V^\dagger$ for every β . For unitaries V, W on $\mathbb{C}^d$ and every operator X on $\mathbb{C}^d \otimes \mathbb{C}^d$,

$$(2.1) \quad \text{Tr}_B[(V \otimes W)X(V \otimes W)^\dagger] = V(\text{Tr}_B X)V^\dagger, \quad \text{Tr}_A[(V \otimes W)X(V \otimes W)^\dagger] = W(\text{Tr}_A X)W^\dagger.$$

Indeed, the (a, a') entry of the first left-hand side is $\sum_{b,x,y,x',y'} V_{ax} W_{by} X_{(x,y),(x',y')} \overline{V_{a'x'}} \overline{W_{by'}}$, and $\sum_b W_{by} \overline{W_{by'}} = \delta_{yy'}$ because $W^\dagger W = 1$; the second identity is the same computation with the roles of the factors exchanged. Let O be the real orthogonal matrix given by Theorem 2 and put $U = (V \otimes V)O(V \otimes V)^\dagger$. Then $U(\tau_\beta(H) \otimes \tau_\beta(H))U^\dagger = (V \otimes V)O(D_\beta \otimes D_\beta)O^\top(V \otimes V)^\dagger$, and by (2.1) both marginals equal $V D_{\beta'} V^\dagger = \tau_{\beta'}(H)$. $\square$

The identities (2.1) are formalised (`partialTraceB_conj_kronecker` and `partialTraceA_conj_kronecker` in `STU.lean`). The reduction itself is also formalised: the theorem `stu_exists_hermitian` (`GeneralH.lean`) states Theorem 1 for every Hermitian H on $\mathbb{C}^d$, with $\tau_\beta(H) = e^{-\beta H} / \text{Tr } e^{-\beta H}$ defined through Mathlib's matrix exponential, and derives it from Theorem 2 via the spectral theorem with sorted eigenvalues, the covariance of τ_β under unitary conjugation, and the local unitary $V \otimes V$. Hence Theorem 1 itself is machine-checked.

The rest of the paper proves Theorem 2. The skeleton of the proof is:

- Proposition 4.4: Theorem 2 holds in dimension d if interval uniformisations exist for all intervals of all boxes of size $n \leq d$ (Lean: `stu_exists_of_intervalUnif`);
- Theorem 3: interval uniformisations exist for every n (Lean: `intervalUnifAll_all`).

3. REDUCTION TO A CLASSICAL PROBLEM

3.1. Majorization. For $x \in \mathbb{R}^\ell$ let $x^\downarrow$ be its nonincreasing rearrangement. Recall that x is *majorized* by λ , written $x \prec \lambda$, if $\sum_i x_i = \sum_i \lambda_i$ and $\sum_{i < k} x_i^\downarrow \leq \sum_{i < k} \lambda_i^\downarrow$ for $1 \leq k \leq \ell$ [15]. We use the following variant, which does not sort λ .

Definition 3.1. For $x, \lambda \in \mathbb{R}^\ell$ we write $x \trianglelefteq \lambda$ if $\sum_{i < \ell} x_i = \sum_{i < \ell} \lambda_i$ and

$$\sum_{i \in S} x_i \leq \sum_{i < |S|} \lambda_i \quad \text{for every } S \subseteq [\ell].$$

Lemma 3.2. *Let $x, x', \lambda, \lambda' \in \mathbb{R}^\ell$.*

- (a) *If $x \leq \lambda$ then $x \prec \lambda$. If λ is nonincreasing, then $x \leq \lambda$ if and only if $x \prec \lambda$.*
- (b) (Superposition.) *If $x \leq \lambda$ and $x' \leq \lambda'$, then $x + x' \leq \lambda + \lambda'$; if $c \geq 0$ then $cx \leq c\lambda$.*
- (c) *If λ is nonincreasing, then $\lambda \leq \lambda$.*
- (d) (0/1 lemma.) *If $0 \leq x_i \leq 1$ for all i and $\sum_i x_i = N \in \{0, \dots, \ell\}$, then $x \leq \mathbf{1}_N$, where $\mathbf{1}_N = (1, \dots, 1, 0, \dots, 0)$ has N ones.*

Proof. (a) If S indexes k largest entries of x , then $\sum_{i < k} x_i^\downarrow = \sum_{i \in S} x_i \leq \sum_{i < k} \lambda_i \leq \sum_{i < k} \lambda_i^\downarrow$. If λ is nonincreasing and $x \prec \lambda$, then for $|S| = k$, $\sum_{i \in S} x_i \leq \sum_{i < k} x_i^\downarrow \leq \sum_{i < k} \lambda_i^\downarrow = \sum_{i < k} \lambda_i$. (b) Add, respectively scale, the defining inequalities. (c) is the case $x = \lambda$ of (a). (d) For every S , $\sum_{i \in S} x_i \leq \min(|S|, N) = \sum_{i < |S|} (\mathbf{1}_N)_i$. $\square$

In the form “if $x^{(m)} \prec \lambda^{(m)}$ and all $\lambda^{(m)}$ are nonincreasing, then $\sum_m x^{(m)} \prec \sum_m \lambda^{(m)}$ ”, part (b) is the superposition principle used throughout. The common ordering matters: $(0, 1) \prec (1, 0)$ and $(0, 1) \prec (0, 1)$, but $(0, 2) \not\prec (1, 1)$.

3.2. Configurations and their realisation. Throughout Sections 3 and 4, $p = (p_0, \dots, p_{d-1})$ satisfies

$$p_0 \geq p_1 \geq \dots \geq p_{d-1} > 0, \quad \sum_i p_i = 1.$$

Definition 3.3. A *configuration* is a real symmetric $d \times d$ matrix $F = (F(i, j))_{i, j \in [d]}$. For $0 \leq s \leq d-1$ its s -th run is $F^{(s)} = (F(i, i+s))_{0 \leq i \leq d-1-s} \in \mathbb{R}^{d-s}$. The runs of $p \otimes p$ are $\lambda^{(s)} = (p_i p_{i+s})_{0 \leq i \leq d-1-s}$; they are nonincreasing. F is *admissible* for p if $F^{(s)} \leq \lambda^{(s)}$ for every s . We put

$$\text{Reach}(p) = \left\{ \left(\sum_j F(i, j) \right)_{i \in [d]} : F \text{ admissible for } p \right\} \subseteq \mathbb{R}^d.$$

The key classical input is the existence direction of the Schur–Horn theorem, due to Horn [18, 10]; a constructive proof by Givens rotations is in [5].

Lemma 3.4 (Horn). *If $x, \lambda \in \mathbb{R}^N$ and $x \prec \lambda$, there is a real orthogonal $N \times N$ matrix G such that the diagonal of $G \text{diag}(\lambda) G^\top$ is x .*

This lemma is formalised in the project (`schurHorn` in `SchurHorn.lean`), with a Chan–Li type induction and the “hockey-stick” form of majorization, $\sum_i \max(x_i - t, 0) \leq \sum_i \max(\lambda_i - t, 0)$ for all $t \in \mathbb{R}$, which needs no sorting.

Proposition 3.5 (realisation). *Let F be admissible for p and $q_i = \sum_j F(i, j)$. There is a real orthogonal matrix O on $\mathbb{R}^{[d] \times [d]}$ such that*

$$\text{Tr}_B [O \text{diag}(p \otimes p) O^\top] = \text{Tr}_A [O \text{diag}(p \otimes p) O^\top] = \text{diag}(q),$$

where $\text{diag}(p \otimes p)$ has entries $p_i p_j$ at (i, j) . The matrix O is block diagonal with respect to the partition of $[d] \times [d]$ into the $2d-1$ diagonals $D_t = \{(i, j) : j - i = t\}$, $|t| \leq d-1$ (the loop $t = 0$, the upper runs $t > 0$ and the lower runs $t < 0$; this is the “tri-run structure” of the Lean development).

Proof. Each diagonal D_t meets every row and every column of the array at most once. For $t = s \geq 0$ the vector $(F(i, j))_{(i, j) \in D_s}$ is the run $F^{(s)}$, and $(p_i p_j)_{(i, j) \in D_s} = \lambda^{(s)}$; for $t = -s < 0$ the two vectors are the same, by the symmetry of F and of $p \otimes p$. By admissibility and Lemma 3.2(a), $F|_{D_t} \prec (p \otimes p)|_{D_t}$, so Lemma 3.4 gives a real orthogonal G_t on $\mathbb{R}^{D_t}$ with $\text{diag}(G_t \text{diag}((p \otimes p)|_{D_t}) G_t^\top) = F|_{D_t}$. Let $O = \bigoplus_t G_t$ and $\rho = O \text{diag}(p \otimes p) O^\top = \bigoplus_t G_t \text{diag}((p \otimes p)|_{D_t}) G_t^\top$. Then $\rho_{(i, j), (i, j)} = F(i, j)$, and $\rho_{(i, j), (i', j')} = 0$ unless (i, j) and (i', j') lie in the same diagonal. In $(\text{Tr}_B \rho)_{a, a'} = \sum_b \rho_{(a, b), (a', b)}$ with $a \neq a'$, the cells (a, b) and (a', b) lie in the same column, hence

in different diagonals, so every term vanishes. Therefore $\text{Tr}_B \rho = \text{diag}(\sum_b F(a, b))_a = \text{diag}(q)$. In the same way $\text{Tr}_A \rho = \text{diag}(\sum_a F(a, b))_b$, which is $\text{diag}(q)$ because F is symmetric. $\square$

Lemma 3.6 (convexity). *Reach(p) is convex.*

Proof. Let F_k be admissible with row sums $q^{(k)}$, and let $\mu_k \geq 0$ with $\sum_k \mu_k = 1$. Then $F = \sum_k \mu_k F_k$ is symmetric with row sums $\sum_k \mu_k q^{(k)}$, and by Lemma 3.2(b), $F^{(s)} = \sum_k \mu_k F_k^{(s)} \leq \sum_k \mu_k \lambda^{(s)} = \lambda^{(s)}$. $\square$

In Lean, admissible configurations and Reach are `ReachTri`, Lemma 3.2 is `SMaj` with `SMaj.add`, `SMaj.smul`, `SMaj.majOn`, `smaj_self`, `smaj_zeroOne`, Proposition 3.5 is `ReachTri.orthogonal` (via `tri_reachable` and `lc_reachable`), and Lemma 3.6 is `ReachTri.convex`; all are in `KirwanFreeCore.lean`.

3.3. Lemma T: thermal targets are mixtures of clipped points.

Lemma 3.7 (Lemma T). *Let $P_0 \geq P_1 \geq \dots \geq P_{d-1} > 0$ and let $g : [0, \infty) \rightarrow \mathbb{R}$ be concave and nondecreasing with $g(0) \geq 0$. There are $\nu_0, \dots, \nu_{d-1} \geq 0$ such that*

$$g(P_i) = \sum_{k=0}^{d-1} \nu_k \min(P_i, P_k) \quad \text{for all } i \in [d].$$

Proof. Let $N_0 = 0$. For $1 \leq k \leq d-1$ let $N_k = \frac{g(P_{k-1}) - g(P_k)}{P_{k-1} - P_k}$ if $P_{k-1} > P_k$, and $N_k = N_{k-1}$ if $P_{k-1} = P_k$. Let $N_d = g(P_{d-1})/P_{d-1}$ and $\nu_k = N_{k+1} - N_k$. Write $\text{sl}(x, y) = \frac{g(y) - g(x)}{y - x}$ for $0 \leq x < y$; concavity gives $\text{sl}(y, z) \leq \text{sl}(x, y)$ for $0 \leq x < y < z$.

Claim 1: for $k \in [d]$ and $0 \leq x < P_k$, $N_k \leq \text{sl}(x, P_k)$. For $k = 0$ this holds because g is nondecreasing. Assume the claim for k , and let $k+1 \leq d-1$ and $0 \leq x < P_{k+1}$. If $P_{k+1} = P_k$ then $N_{k+1} = N_k \leq \text{sl}(x, P_k) = \text{sl}(x, P_{k+1})$. Otherwise $N_{k+1} = \text{sl}(P_{k+1}, P_k) \leq \text{sl}(x, P_{k+1})$ by concavity.

Claim 2: $\nu_k \geq 0$. For $k \leq d-2$: if $P_{k+1} = P_k$ then $\nu_k = 0$; otherwise Claim 1 with $x = P_{k+1}$ gives $N_k \leq \text{sl}(P_{k+1}, P_k) = N_{k+1}$. For $k = d-1$, Claim 1 with $x = 0$ gives $N_{d-1} \leq (g(P_{d-1}) - g(0))/P_{d-1} \leq N_d$, since $g(0) \geq 0$.

Claim 3: the identity. In all cases $N_{k+1}(P_k - P_{k+1}) = g(P_k) - g(P_{k+1})$ for $k \leq d-2$, and $N_d P_{d-1} = g(P_{d-1})$. Since $\min(P_i, P_k) = P_i$ for $k < i$ and $= P_k$ for $k \geq i$, and $\sum_{k < i} \nu_k = N_i$,

$$\begin{aligned} \sum_k \nu_k \min(P_i, P_k) &= N_i P_i + \sum_{k=i}^{d-1} (N_{k+1} - N_k) P_k \\ &= \sum_{k=i}^{d-2} N_{k+1} (P_k - P_{k+1}) + N_d P_{d-1} = g(P_i). \end{aligned} \quad \square$$

Corollary 3.8 (thermal targets). *Let $E_0 \leq \dots \leq E_{d-1}$, $\beta > 0$, $0 \leq \beta' \leq \beta$, $p = \gamma_\beta(E)$ and $q = \gamma_{\beta'}(E)$. For $k \in [d]$ put*

$$Z_k = \sum_i \min(p_i, p_k) > 0, \quad w_k = \min(p, p_k)/Z_k,$$

the clipped points of p . Then $q = \sum_k \mu_k w_k$ with $\mu_k \geq 0$ and $\sum_k \mu_k = 1$.

Proof. p is positive and nonincreasing. With $t = \beta'/\beta \in [0, 1]$ we have $p_i^t = e^{-\beta' E_i} / (\sum_j e^{-\beta E_j})^t$, hence $q_i = p_i^t / S$ with $S = \sum_j p_j^t$. The function $g(x) = x^t$ (with $g \equiv 1$ if $t = 0$) is concave and nondecreasing on $[0, \infty)$ with $g(0) \geq 0$. Lemma 3.7 gives $\nu_k \geq 0$ with $p_i^t = \sum_k \nu_k \min(p_i, p_k)$. Put $\mu_k = \nu_k Z_k / S \geq 0$. Then $\sum_k \mu_k w_k = q$, and $\sum_k \mu_k = \sum_k \nu_k \sum_i \min(p_i, p_k) / S = \sum_i p_i^t / S = 1$. $\square$

In Lean these are `lemmaT` and `thermal_clip_weights` (`KirwanFreeCore.lean`). By Lemma 3.6 and Proposition 3.5, Theorem 2 follows once every clipped point w_k lies in $\text{Reach}(p)$. This is the content of Proposition 4.3 below.

3.4. The decomposition of $p \otimes p$. Fix $k \in [d]$. Put $p_d = 0$ and

$$c_m = p_m - p_{m+1} \quad (m \in [d]), \quad B_i = \min(p_i, p_k), \quad Z = \sum_i B_i = Z_k.$$

Then $c_m \geq 0$ and $c_{d-1} = p_{d-1} > 0$.

Lemma 3.9. *For all $i \in [d]$:*

$$\begin{aligned} (I1) \quad p_i &= \sum_{M=0}^{d-1} \mathbf{1}[i \leq M] c_M, & (I2) \quad B_i &= \sum_{M=k}^{d-1} \mathbf{1}[i \leq M] c_M, \\ (I3) \quad p_i - B_i &= \sum_{m=0}^{k-1} \mathbf{1}[i \leq m] c_m, & (I4) \quad Z &= \sum_{M=k}^{d-1} (M+1) c_M, \\ (I5) \quad 1 - Z &= \sum_{m=0}^{k-1} (m+1) c_m. \end{aligned}$$

Proof. (I1) telescopes. For (I2), the right-hand side is $p_{\max(i,k)} = \min(p_i, p_k)$ because p is nonincreasing. (I3) is (I1)–(I2). Summing (I2) and (I3) over i gives (I4) and (I5), using $\sum_i p_i = 1$. $\square$

For $u, v \in \mathbb{N}$ let $C(u, v) = ([0, u] \times [0, v]) \cup ([0, v] \times [0, u]) \subseteq \mathbb{N}^2$ be the *cross* with arms u, v , and $\chi_{u,v}$ its indicator function. It is symmetric in (u, v) and in its two arguments, and $Q_u = \chi_{u,u}$ is the indicator of the square $[0, u]^2$. Crosses and squares are symmetric Young diagrams (down-closed subsets of $\mathbb{N}^2$). (In the working notes accompanying the Lean code the squares are called Q and the crosses U ; in Lean, $\chi_{u,v}$ is `crossInd u v`.)

Lemma 3.10 (decomposition). *Let $\omega_{m,M} = \frac{1}{2} c_m c_M (1 + \mathbf{1}[M \geq k])$ for $m < k$ and $M \in [d]$. For all $i, j \in [d]$,*

$$p_i p_j = B_i B_j + \sum_{m=0}^{k-1} \sum_{M=0}^{d-1} \omega_{m,M} (Q_{\min(m,M)}(i, j) + \chi_{m,M}(i, j)).$$

Proof. By inclusion–exclusion, $Q_{\min(m,M)}(i, j) + \chi_{m,M}(i, j) = \mathbf{1}[i \leq m] \mathbf{1}[j \leq M] + \mathbf{1}[i \leq M] \mathbf{1}[j \leq m]$. By (I1) and (I2), $\sum_M \frac{1}{2} c_M (1 + \mathbf{1}[M \geq k]) \mathbf{1}[j \leq M] = \frac{1}{2} (p_j + B_j)$. With (I3) the double sum becomes $\frac{1}{2} (p_i - B_i)(p_j + B_j) + \frac{1}{2} (p_j - B_j)(p_i + B_i) = p_i p_j - B_i B_j$. $\square$

Lemma 3.11 (runs of a cross). *Let $u \leq v$ and $s \geq 0$. For $i \in \mathbb{N}$, $\chi_{u,v}(i, i+s) = 1$ if and only if $s \leq v$ and $i \leq \min(u, v-s)$. Consequently, for $v < d$ the run $(\chi_{u,v}(i, i+s))_{i < d-s}$ equals $\mathbf{1}_{N_s}$ with $N_s = \#\{a \in [v-u, v] : a \geq s\}$, and $C(u, v) \cap [d]^2$ has $\mathbf{m}(v-u, v)$ cells, where*

$$\mathbf{m}(\ell, h) = \sum_{a=\ell}^h (2a+1) = (h+1)^2 - \ell^2.$$

Proof. The condition $(i \leq v \wedge i+s \leq u)$ implies $(i \leq u \wedge i+s \leq v)$ because $u \leq v$, so $\chi_{u,v}(i, i+s) = 1$ iff $i \leq u$ and $i+s \leq v$. For $s \leq v$ this gives $\min(u, v-s)+1 = \#\{a \in [v-u, v] : a \geq s\}$ ones at the start of the run; for $s > v$ both sides are 0. The cell count is $(u+1)(v+1) + (v+1)(u+1) - (u+1)^2 = (v+1)^2 - (v-u)^2$. $\square$

Thus $C(u, v)$ is the symmetric Young diagram whose Frobenius hooks have the arm lengths $v-u, \dots, v$: its runs only depend on the *arm interval* $[v-u, v]$. The square Q_u has arm interval $[0, u]$.

3.5. The Hall identity. For $k \leq M \leq d-1$ define the *capacity* of the box $[0, M]$, the mass of the *big crosses* assigned to it, and the *leftover*:

$$A_M = \frac{(1-Z^2)(M+1)c_M}{Z}, \quad \Lambda_M = \sum_{m=0}^{k-1} \omega_{m,M} \mathbf{m}(M-m, M), \quad L_M = A_M - \Lambda_M,$$

and $L = \sum_{M=k}^{d-1} L_M$.

Lemma 3.12 (Hall identity). *For $k \leq M \leq d-1$,*

$$L_M = c_M \left(\frac{(M+1)(1-Z)^2}{Z} + S_k \right), \quad S_k = \sum_{i=0}^{k-1} (2i+1)(p_i - p_k) \geq 0.$$

In particular $L_M \geq 0$. Moreover $\sum_{M=k}^{d-1} A_M = 1 - Z^2$, and $L > 0$ unless $p_0 = p_1 = \dots = p_k$.

Proof. For $M \geq k$ we have $\omega_{m,M} = c_m c_M$ and $\mathbf{m}(M-m, M) = (m+1)(2M+1-m) = 2(m+1)(M+1) - (m+1)^2$. By (I5) and the summation by parts $\sum_{m < k} (m+1)^2 c_m = \sum_{m < k} \sum_{i \leq m} (2i+1) c_m = S_k$, we get $\Lambda_M = c_M(2(M+1)(1-Z) - S_k)$. Hence

$$L_M = c_M \left((M+1)(1-Z) \left(\frac{1+Z}{Z} - 2 \right) + S_k \right) = c_M \left(\frac{(M+1)(1-Z)^2}{Z} + S_k \right).$$

By (I4), $\sum_{M \geq k} A_M = (1-Z^2)Z/Z$. If $p_m > p_{m+1}$ for some $m < k$, then $1-Z \geq (m+1)c_m > 0$ by (I5), so $L \geq L_{d-1} \geq d c_{d-1} (1-Z)^2/Z > 0$. $\square$

The Lean development proves the inequality $L_M \geq (M+1)c_M(1-Z)^2/Z$, which suffices (`leftM_nonneg`, `Ltot_pos`, `sum_aM` in `KirwanFree.lean`, using the bound $\mathbf{m}(M-m, M) \leq 2(m+1)(M+1)$); Lemma 3.10 is `wk_cell_identity` there.

4. INTERVAL UNIFORMISATIONS

4.1. Definition. The following definition is, clause by clause, the Lean structure `IsIntervalUnif n lo hi u` (`IntervalUnif.lean`), with $\ell = \text{lo}$ and $h = \text{hi}$.

Definition 4.1 (interval uniformisation). Let $n \geq 1$ and $0 \leq \ell \leq h \leq n-1$. An *interval uniformisation* of $[\ell, h]$ in the n -box is a function $Y : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ such that

- (U1) $Y(i, j) = Y(j, i)$ for all i, j ;
- (U2) $0 \leq Y(i, j) \leq 1$ for all i, j ;
- (U3) $Y(i, j) = 0$ whenever $i \geq n$;
- (U4) $\sum_{i=0}^{n-1} Y(i, i+s) = r_s := \#\{a \in [\ell, h] : a \geq s\}$ for every $s \geq 0$;
- (U5) $\sum_{j=0}^{n-1} Y(r, j) = \frac{1}{n} \sum_{a=\ell}^h (2a+1) = \frac{\mathbf{m}(\ell, h)}{n}$ for every $r \in [n]$.

We say that `UNIF`(n) holds (Lean: `IntervalUnifAll n`) if such a Y exists for every $0 \leq \ell \leq h \leq n-1$.

By (U1) and (U3), Y vanishes outside $[n]^2$. The runs of Y have the totals of the runs of a symmetric Young diagram with arm interval $[\ell, h]$, but all row sums are equal.

Lemma 4.2. *Let Y be an interval uniformisation of $[v-u, v]$ in the n -box, where $u \leq v < n \leq d$. Then for every $s < d$ the run $(Y(i, i+s))_{i < d-s}$ satisfies $(Y(i, i+s))_{i < d-s} \trianglelefteq (\chi_{u,v}(i, i+s))_{i < d-s}$, and for $r \in [d]$, $\sum_{j < d} Y(r, j) = \mathbf{1}[r < n] \mathbf{m}(v-u, v)/n$.*

Proof. The entries lie in $[0, 1]$ by (U2). Since $Y(i, i+s) = 0$ when $i+s \geq n$ (by (U1), (U3)) and $n \leq d$, both $\sum_{i < d-s} Y(i, i+s)$ and the sum in (U4) reduce to $\sum_{i < n-s} Y(i, i+s)$, so the run sums to r_s . By Lemma 3.11 the cross run is $\mathbf{1}_{r_s}$, and Lemma 3.2(d) applies. The row sums follow from (U3) and (U5). $\square$

In Lean this is `IsIntervalUnif.smaj` and `IsIntervalUnif.row'`.

4.2. Clipped points are reachable.

Proposition 4.3. *Assume that for every $n \leq d$ and every $0 \leq \ell \leq h \leq n-1$ an interval uniformisation $Y_n[\ell, h]$ of $[\ell, h]$ in the n -box is given. Then $w_k = B/Z \in \text{Reach}(p)$ for every $k \in [d]$.*

Proof. Fix k and use the notation of Section 3. All matrices below are restricted to $[d] \times [d]$.

Trivial case. If $p_0 = \dots = p_k$, then $B = p$, $Z = 1$ and $w_k = p$; the configuration $F(i, j) = p_i p_j$ is admissible by Lemma 3.2(c) and has row sums p .

General case. Otherwise $L > 0$ by Lemma 3.12. Put $\xi_M = L_M/L$ for $k \leq M \leq d-1$, so $\xi_M \geq 0$ and $\sum_M \xi_M = 1$. For $0 \leq \ell \leq h \leq k-1$ define the *flexible piece*

$$Y^{\text{fl}}[\ell, h] = \sum_{M=k}^{d-1} \xi_M Y_{M+1}[\ell, h],$$

a convex combination of uniformisations in the boxes $[0, M]$, $M \geq k > h$. For $m < k$ and $M \in [d]$ let

$$G_{m,M} = \begin{cases} Y_{M+1}[M-m, M] + Y^{\text{fl}}[0, m], & M \geq k, \\ Y^{\text{fl}}[|m-M|, \max(m, M)] + Y^{\text{fl}}[0, \min(m, M)], & M < k, \end{cases}$$

and

$$F = B \otimes B + \sum_{m=0}^{k-1} \sum_{M=0}^{d-1} \omega_{m,M} G_{m,M}.$$

Thus the square $Q_{\min(m,M)}$ is always replaced by a flexible piece; a cross $\chi_{m,M}$ with $M \geq k$ (a *big cross*) is replaced by a uniformisation of its arm interval $[M-m, M]$ in its own box $[0, M]$ (then $\min(m, M) = m$); and a cross with $M < k$ is replaced by a flexible piece.

Symmetry. Every term is symmetric.

Runs. Fix $s < d$. The run of $B \otimes B$ is nonincreasing, so it is $\leq$ itself (Lemma 3.2(c)). By Lemma 4.2 (with $n = M+1 \leq d$), the run of $Y_{M+1}[M-m, M]$ is $\leq$ the run of $\chi_{m,M}$, and for $v < k \leq M$ the run of $Y_{M+1}[v-u, v]$ is $\leq$ the run of $\chi_{u,v}$; averaging with the weights ξ_M (Lemma 3.2(b)), the run of $Y^{\text{fl}}[v-u, v]$ is $\leq$ the run of $\chi_{u,v}$. Hence the run of $G_{m,M}$ is $\leq$ the run of $Q_{\min(m,M)} + \chi_{m,M}$. Adding with the weights $\omega_{m,M} \geq 0$ and using Lemma 3.10, $F^{(s)} \leq \lambda^{(s)}$.

Rows. Fix $r \in [d]$ and let $\theta(r) = \sum_{M=k}^{d-1} \xi_M \mathbf{1}[r \leq M]/(M+1)$. By Lemma 4.2, row r of $Y_{M+1}[\ell, h]$ sums to $\mathbf{m}(\ell, h) \mathbf{1}[r \leq M]/(M+1)$ and row r of $Y^{\text{fl}}[\ell, h]$ to $\mathbf{m}(\ell, h) \theta(r)$. Row r of $B \otimes B$ sums to $B_r Z$. Collecting terms,

$$\sum_j F(r, j) = B_r Z + \sum_{M=k}^{d-1} \Lambda_M \frac{\mathbf{1}[r \leq M]}{M+1} + \Lambda^{\text{fl}} \theta(r),$$

where Λ^{fl} is the total ω -weighted mass of the flexible pieces. The total mass of all pieces is $\sum_{m,M} \omega_{m,M} (\mathbf{m}(0, \min(m, M)) + \mathbf{m}(|m-M|, \max(m, M))) = \sum_{i,j < d} (p_i p_j - B_i B_j) = 1 - Z^2$ by Lemmas 3.10 and 3.11, so $\Lambda^{\text{fl}} = 1 - Z^2 - \sum_M \Lambda_M = \sum_M (A_M - \Lambda_M) = L$ by Lemma 3.12. Since $\xi_M L = L_M$,

$$\begin{aligned} \sum_j F(r, j) &= B_r Z + \sum_{M=k}^{d-1} (\Lambda_M + L_M) \frac{\mathbf{1}[r \leq M]}{M+1} = B_r Z + \frac{1-Z^2}{Z} \sum_{M=k}^{d-1} \mathbf{1}[r \leq M] c_M \\ &= B_r Z + \frac{(1-Z^2)B_r}{Z} = \frac{B_r}{Z}, \end{aligned}$$

using the definition of A_M and (I2). Hence F is admissible with row sums w_k . $\square$

In Lean this is `wkU_reach` (with `wkU_smaj` and `wkU_rows`) in `KirwanFreeU.lean`. The boxes used have size $n = M + 1 \leq d$; the big crosses need the intervals $[M - m, M]$ with $h = n - 1$, and the flexible pieces need intervals with $h \leq n - 2$.

4.3. The reduction and the main theorem.

Proposition 4.4 (reduction). *If $\text{UNIF}(n)$ holds for every $n \leq d$, then Theorem 2 holds in dimension d .*

Proof. For $d = 0$ there is nothing to prove. For $d \geq 1$ let $p = \gamma_\beta(E)$ and $q = \gamma_{\beta'}(E)$. By Corollary 3.8, q is a convex combination of the w_k , which lie in $\text{Reach}(p)$ by Proposition 4.3. By Lemma 3.6, $q \in \text{Reach}(p)$, and Proposition 3.5 gives a real orthogonal O with both marginals of $O \text{diag}(p \otimes p) O^\top$ equal to $\text{diag}(q)$. Since $D_\beta \otimes D_\beta = \text{diag}(p \otimes p)$ and $D_{\beta'} = \text{diag}(q)$, the matrix $U = O$ satisfies Theorem 2. $\square$

This is `stu_exists_of_intervalUnif` (`KirwanFreeU.lean`). The remaining ingredient is:

Theorem 3 (interval uniformisations exist). *$\text{UNIF}(n)$ holds for every $n \geq 1$.*

Proof of Theorem 2. Combine Proposition 4.4 and Theorem 3. (In Lean: `stu_exists_unconditional`, in `KirwanFreeFinal.lean`.) $\square$

Theorem 3 is proved in Section 9.4. The constructions are summarised in Table 1, where $K = n - 1 - h$ and $M = n - \ell$.

interval	case	construction	reference
$[0, n - 1]$	(A) $h = n - 1$	J_n (all ones)	Lemma 4.6
$[\ell, n - 1], \ell \geq 1$	(A) $h = n - 1$	$J_n - \mathcal{R}_n^{\ell-1}$	Lemma 4.6
$[0, h], h \leq n - 2$	(B) $\ell = 0$	renewal square $\mathcal{R}_n^h$	Proposition 4.5
$[\ell, h], 1 \leq \ell \leq h \leq n - 2$	(C) $n \leq 2K + 1$	Y^0 (no dipole)	Proposition 9.1
same	(D) $K = 1, n \geq 4$	$Y^{2/7}$	Proposition 9.5
same	(E) $K \geq 2, n \geq 2K + 2$	Y^{α^*}	Proposition 9.4

TABLE 1. The interval uniformisations. Y^α is the T/H/G matrix with one dipole of amplitude α (Definition 6.1); α^* is given in (8.2).

4.4. Renewal squares and complements.

Proposition 4.5 (renewal squares). *Let $n \geq 2$, $0 \leq m \leq n - 2$ and $K = n - m - 1 \geq 1$. Define $\eta_\ell = 1 - \frac{K^2}{(n-\ell)(n-\ell+1)}$ for $1 \leq \ell \leq m + 1$ and $\eta_\ell = 0$ for $\ell = 0$ and for $\ell \geq m + 2$; put $A_r = \sum_{t=0}^{m+1} (-1)^t \eta_{r+t}$ for $r \geq 0$, and*

$$\mathcal{R}_n^m(i, j) = A_{|i-j|+1} + A_{\min(i+j+2, 2n-i-j)} \quad (i, j < n), \quad \mathcal{R}_n^m(i, j) = 0 \text{ otherwise.}$$

Then $\mathcal{R}_n^m$ is an interval uniformisation of $[0, m]$ in the n -box.

Proof. (i) *Tails.* In the sum $A_r + A_{r+1}$ all terms cancel in pairs except the first term of A_r and the last term of A_{r+1} , so $A_r + A_{r+1} = \eta_r + (-1)^{m+1} \eta_{r+m+2} = \eta_r$, because $r + m + 2 \geq m + 2$.

(ii) *Weights.* For $1 \leq \ell \leq m$, $\eta_\ell - \eta_{\ell+1} = \frac{2K^2}{(n-\ell-1)(n-\ell)(n-\ell+1)} \geq 0$, and $\eta_{m+1} = \frac{1}{K+1}$. Since $(n-\ell)(n-\ell+1) \geq K(K+1) > K^2$ for $\ell \leq m+1$, we get $1 \geq \eta_1 \geq \eta_2 \geq \dots \geq \eta_{m+1} > 0 = \eta_{m+2} = \eta_{m+3} = \dots$.

(iii) *Alternating sums.* For $r \geq 1$ and $\kappa \geq 0$, (i) gives $A_r + A_{r+2\kappa+1} = \sum_{t=0}^{2\kappa} (-1)^t \eta_{r+t}$, an alternating sum of a nonincreasing nonnegative sequence; hence $0 \leq A_r + A_{r+2\kappa+1} \leq \eta_r \leq 1$ (by induction on κ : $A_r + A_{r+2\kappa+3} = (\eta_r - \eta_{r+1}) + (A_{r+2} + A_{r+2\kappa+3})$).

(iv) *Entries.* For $i, j < n$ one checks $\min(i + j + 2, 2n - i - j) = |i - j| + 2 + 2\kappa$ with $\kappa = \min(i, j, n - 1 - i, n - 1 - j)$. So $\mathcal{R}_n^m(i, j) = A_{s+1} + A_{s+1+2\kappa+1} \in [0, \eta_{s+1}] \subseteq [0, 1]$ with $s = |i - j|$, and $\mathcal{R}_n^m(i, j) = 0$ if $s \geq m + 1$, since then both indices are $\geq m + 2$ and $A_r = 0$ for $r \geq m + 2$. (U1)–(U3) hold.

(v) *Rows.* Row 0: $\mathcal{R}_n^m(0, j) = A_{j+1} + A_{j+2} = \eta_{j+1}$, so it sums to $\sum_{\ell=1}^{m+1} \eta_\ell = (m + 1) - K^2(\frac{1}{n-m-1} - \frac{1}{n}) = \frac{(m+1)^2}{n} = \frac{m(0, m)}{n}$. For $0 \leq r \leq n - 2$, substituting $j \mapsto j + 1$ shows that the Toeplitz part changes by $\sum_j A_{|r+1-j|+1} - \sum_j A_{|r-j|+1} = A_{r+2} - A_{n-r}$, and the Hankel part by $\sum_j A_{\varphi(r+1+j)} - \sum_j A_{\varphi(r+j)} = A_{\varphi(r+n)} - A_{\varphi(r)} = A_{n-r} - A_{r+2}$, where $\varphi(S) = \min(S + 2, 2n - S)$ (all sums over $0 \leq j \leq n - 1$). So all rows have the same sum. This is (U5).

(vi) *Runs.* Let $W_\ell = 2\eta_\ell + (n - \ell - 1)(\eta_\ell - \eta_{\ell+1})$. By (ii), $W_\ell = 2$ for $1 \leq \ell \leq m$ and $W_{m+1} = (n - m)\eta_{m+1} = 1$. For $0 \leq s \leq n - 2$ we claim $\text{run}_s = W_{s+1} + \text{run}_{s+2}$, where $\text{run}_s = \sum_i \mathcal{R}_n^m(i, i + s)$. The first and last cells of run s , $(0, s)$ and $(n - 1 - s, n - 1)$, have $\kappa = 0$ and value $A_{s+1} + A_{s+2} = \eta_{s+1}$. For $0 \leq i \leq n - s - 3$, the cells $(i + 1, i + 1 + s)$ and $(i, i + s + 2)$ have the same $i + j$, hence the same Hankel term, and Toeplitz terms A_{s+1} and A_{s+3} , which differ by $\eta_{s+1} - \eta_{s+2}$ by (i). The cells $(i, i + s + 2)$, $0 \leq i \leq n - s - 3$, form run $s + 2$. Summing, $\text{run}_s = 2\eta_{s+1} + \text{run}_{s+2} + (n - s - 2)(\eta_{s+1} - \eta_{s+2}) = W_{s+1} + \text{run}_{s+2}$. Now $\text{run}_s = 0$ for $s \geq m + 1$ by (iv), $\text{run}_m = W_{m+1} = 1$, $\text{run}_{m-1} = W_m = 2$ if $m \geq 1$, and downward induction gives $\text{run}_s = m + 1 - s = \#\{a \in [0, m] : a \geq s\}$ for $s \leq m$. This is (U4). $\square$

The name comes from an equivalent description: $\mathcal{R}_n^m$ is the superposition of the reversal permutation matrices of the blocks $[0, \ell - 1]$ and $[n - \ell, n - 1]$ with weight η_ℓ and of the interior blocks $[a, a + \ell - 1]$ ($1 \leq a \leq n - 1 - \ell$) with weight $\eta_\ell - \eta_{\ell+1}$, a stationary renewal process of block boundaries. We do not need this description (it was checked for $n \leq 24$, Appendix B). In Lean, $\mathcal{R}_n^m$ is `renU` and the proposition is `isIntervalUnif_renU` (`Renewal.lean`).

Lemma 4.6 (*J and complements*). *The all-ones matrix J_n of the n -box is an interval uniformisation of $[0, n - 1]$. If Y is an interval uniformisation of $[0, m]$ in the n -box, $m \leq n - 2$, then $J_n - Y$ (restricted to the box) is one of $[m + 1, n - 1]$.*

Proof. For J_n : $\text{run}_s = n - s = \#\{a \in [0, n - 1] : a \geq s\}$ and all rows equal $n = n^2/n$. For $J_n - Y$: entries lie in $[0, 1]$; $\text{run}_s = (n - s) - \#\{a \in [0, m] : a \geq s\} = \#\{a \in [m + 1, n - 1] : a \geq s\}$; rows equal $n - (m + 1)^2/n = m(m + 1, n - 1)/n$. $\square$

In Lean: `isIntervalUnif_boxJ` and `IsIntervalUnif.compl` (`Renewal.lean`).

5. T/H/G MATRICES

To avoid confusion with the Hamiltonian, the three sequences below are written in sans-serif.

Definition 5.1. Let $n \geq 1$ and $T, H, G : \mathbb{N} \rightarrow \mathbb{R}$. The *T/H/G matrix* $Y = Y[T, H, G]$ of the n -box is defined for $i, j < n$ by

$$Y(i, j) = \begin{cases} \frac{1}{2}(T(|i - j|) + H(m'(i, j))), & i + j \neq n - 1, \\ \frac{1}{2}(T(|i - j|) + G(|i - j|)), & i + j = n - 1, \end{cases} \quad m'(i, j) = \min(i + j + 1, 2n - 1 - i - j),$$

and $Y(i, j) = 0$ if $i \geq n$ or $j \geq n$. We write $\text{run}_s = \sum_{i < n} Y(i, i + s)$ and $\text{row}_r = \sum_{j < n} Y(r, j)$.

The Toeplitz part depends on $|i - j|$, the Hankel part on $i + j$ (through the “level” m'), and the antidiagonal $i + j = n - 1$ is treated separately.

Lemma 5.2 (*levels and cells*). *Let $i, j < n$ and $s = |i - j|$. Then $m'(i, j) - s$ is odd and positive, $m'(i, j) \leq n$, and $m'(i, j) = n$ if and only if $i + j = n - 1$; if $i + j = n - 1$ then $n - s$ is odd.*

Consequently Y is symmetric, and $0 \leq Y \leq 1$ holds provided

$$(5.1) \quad 0 \leq T(s) + H(u) \leq 2 \quad \text{for all } 0 \leq s < u \leq n-1 \text{ with } u-s \text{ odd,}$$

$$(5.2) \quad 0 \leq T(s) + G(s) \leq 2 \quad \text{for all } 0 \leq s \leq n-1 \text{ with } n-s \text{ odd.}$$

Proof. $i+j+1-s = 2\min(i, j) + 1$ and $2n-1-i-j-s = 2n-1-2\max(i, j) \geq 1$ are odd and positive. If $i+j \leq n-2$ then $m' \leq i+j+1 \leq n-1$; if $i+j \geq n$ then $m' \leq 2n-1-i-j \leq n-1$; if $i+j = n-1$ then $m' = n$ and $n-s = 2\min(i, j) + 1$. $\square$

Lemma 5.3 (runs). (a) $\text{run}_s = 0$ for $s \geq n$. (b) $\text{run}_{n-1} = \frac{1}{2}(T(n-1) + G(n-1))$. (c) If $n \geq 2$, $\text{run}_{n-2} = T(n-2) + H(n-1)$. (d) For $0 \leq s \leq n-3$,

$$\begin{aligned} & 2\text{run}_s - 2\text{run}_{s+2} \\ &= (n-s)T(s) - (n-s-2)T(s+2) + 2H(s+1) + \mathbf{1}[n-s \text{ odd}](G(s) - G(s+2)). \end{aligned}$$

Proof. (a) is clear. (b) Run $n-1$ is the antidiagonal cell $(0, n-1)$. (c) Run $n-2$ consists of $(0, n-2)$ and $(1, n-1)$, with $i+j \in \{n-2, n\}$ and $m' = n-1$ in both cases. (d) Let $0 \leq i \leq n-s-3$. The cells $(i+1, i+1+s)$ and $(i, i+s+2)$ have the same sum $i+j = 2i+s+2$, so both are antidiagonal or neither is, and they have the same level m' . Their difference is $\frac{1}{2}(T(s) - T(s+2))$, plus $\frac{1}{2}(G(s) - G(s+2))$ when $2i+s+3 = n$; such an i exists (and is unique) iff $n-s$ is odd. The cells $(i, i+s+2)$, $0 \leq i \leq n-s-3$, form run $s+2$. The two remaining cells of run s , $(0, s)$ and $(n-1-s, n-1)$, are not antidiagonal and have level $s+1$, so each equals $\frac{1}{2}(T(s) + H(s+1))$. Summing and multiplying by 2 gives the claim. $\square$

Lemma 5.4 (row shift). For $0 \leq r \leq n-2$,

$$\begin{aligned} 2\text{row}_{r+1} - 2\text{row}_r &= (T(r+1) - H(r+1)) - (T(n-1-r) - H(n-1-r)) \\ &\quad + G(|2r+3-n|) - G(|2r+1-n|). \end{aligned}$$

Proof. Let $H^*(S) = H(\min(S+1, 2n-1-S))$ for $S \neq n-1$ and $H^*(n-1) = 0$. The antidiagonal cell of row r is $j = n-1-r$, at distance $|2r+1-n|$, so $2\text{row}_r = \sum_{j < n} T(|r-j|) + \sum_{j < n} H^*(r+j) + G(|2r+1-n|)$. Shifting r to $r+1$, the Toeplitz sum gains $T(r+1)$ (the term $j=0$) and loses $T(n-1-r)$ (the old term $j=n-1$); the Hankel sum $\sum_{S=r+1}^{r+n} H^*(S)$ loses $H^*(r) = H(r+1)$ and gains $H^*(r+n) = H(n-1-r)$; the antidiagonal term changes from $G(|2r+1-n|)$ to $G(|2r+3-n|)$. $\square$

Lemma 5.5 (the row value is forced). Let $Y : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ satisfy (U1), (U3) and (U4) for $[\ell, h]$, $h < n$, and suppose all rows $r \in [n]$ have the same sum c . Then $c = m(\ell, h)/n$, i.e. (U5) holds.

Proof. Splitting $[n]^2$ into the upper runs and the strictly lower runs and using symmetry, $nc = \sum_{s \geq 0} \text{run}_s + \sum_{s \geq 1} \text{run}_s = \sum_{s \geq 0} (r_s + r_{s+1}) = \sum_{a=\ell}^h ((a+1) + a)$. $\square$

In Lean these are `thgU`, `thg_cells`, `thg_run_rec`, `thg_run_top`, `thg_run_top2`, `thg_row_succ` and `row_const_value` (`DipoleUnif.lean`).

6. THE ONE-DIPOLE CONSTRUCTION

Definition 6.1. Let $1 \leq \ell \leq h \leq n-2$ and $\alpha \in \mathbb{R}$, and put $K = n-1-h$ and $M = n-\ell$, so that $K \geq 1$ and $K+1 \leq M \leq n-1$. For integers $x \geq 0$ and $L \geq 0$ let

$$q_L(x) = \mathbf{1}[x \geq L+1](2x(x-1)-2L^2), \quad Q(x) = q_K(x) - q_M(x) + 2(K+1)\alpha \mathbf{1}[K+2 \leq x \leq 2K+2],$$

and define $\vartheta : \mathbb{N} \rightarrow \mathbb{R}$ by $\vartheta(0) = \vartheta(1) = 0$ and

$$\vartheta(x) = \frac{Q(x)}{x(x-1)} - \vartheta(x-1) \quad (x \geq 2).$$

Set $T(s) = \vartheta(n-s)$ for $0 \leq s \leq n$ and $T(s) = 0$ for $s > n$,

$$H(u) = T(u) - \alpha \mathbf{1}[u = h], \quad G(u) = \alpha \mathbf{1}[u \leq 2h - n - 1],$$

and let Y^α be the $T/H/G$ matrix $Y[T, H, G]$ of the n -box.

Here $x = n - s$ is the natural coordinate: ϑ depends only on K , M and α , not on n , which only truncates the range to $x \leq n$. The pair $(-\alpha$ at the Hankel level h , $+\alpha$ on the antidiagonal cells with $|i - j| \leq 2h - n - 1$) is the *dipole*; for $\alpha = 0$ the construction is the difference $\mathcal{R}_n^h - \mathcal{R}_n^{\ell-1}$ of two renewal squares (Remark 8.2). In Lean, $\mathcal{Q}$, ϑ and Y^α are `dipQ`, `dipTau` and `dipU`; Lean's definition $\vartheta(x+1) = \mathcal{Q}(x+1)/((x+1)x) - \vartheta(x)$ for all $x \geq 0$ gives the same values, because Lean divides by zero to zero at $x = 0$.

Lemma 6.2. *For every $x \geq 0$,*

$$\mathcal{Q}(x+1) - \mathcal{Q}(x) = x \left(2\mathbf{1}[x \geq K] + 2\mathbf{1}[x \geq K+1] - 2\mathbf{1}[x \geq M] - 2\mathbf{1}[x \geq M+1] + 2\alpha \mathbf{1}[x = K+1] - \alpha \mathbf{1}[x = 2K+2] \right).$$

Proof. $q_L(x+1) - q_L(x)$ equals $4x$ if $x \geq L+1$, $2L = 2x$ if $x = L$, and 0 if $x < L$; that is, $x(2\mathbf{1}[x \geq L] + 2\mathbf{1}[x \geq L+1])$. The window term changes by $+2(K+1)\alpha = 2\alpha x$ at $x = K+1$ and by $-2(K+1)\alpha = -\alpha x$ at $x = 2K+2$. $\square$

Proposition 6.3. *In the setting of Definition 6.1:*

- (a) *for every α , $\text{run}_s(Y^\alpha) = r_s = \#\{a \in [\ell, h] : a \geq s\}$ for all $s \geq 0$;*
- (b) *if $\alpha = 0$ or $2h \geq n$, all rows of Y^α have sum $\mathbf{m}(\ell, h)/n$;*
- (c) *if moreover*

$$(\text{CELL1}) \quad 0 \leq \vartheta(x) + \vartheta(y) - \alpha \mathbf{1}[y = K+1] \leq 2 \quad \text{for all } 1 \leq y < x \leq n \text{ with } x - y \text{ odd},$$

$$(\text{CELL2}) \quad 0 \leq \vartheta(x) + \alpha \mathbf{1}[x \geq 2K+3] \leq 2 \quad \text{for all odd } x \text{ with } 1 \leq x \leq n,$$

then Y^α is an interval uniformisation of $[\ell, h]$ in the n -box.

Proof. Note $n \geq 3$. (a) By Lemma 5.3(a), $\text{run}_s = 0 = r_s$ for $s \geq n$. Next, $\text{run}_{n-1} = \frac{1}{2}(\vartheta(1) + \alpha \mathbf{1}[n-1 \leq 2h-n-1]) = 0 = r_{n-1}$, because $h \leq n-2$. And $\text{run}_{n-2} = \vartheta(2) + \vartheta(1) - \alpha \mathbf{1}[n-1 = h] = \vartheta(2) = \mathcal{Q}(2)/2 = \mathbf{1}[K=1] = \mathbf{1}[h=n-2] = r_{n-2}$, since $\mathcal{Q}(2) = q_K(2) = 2$ if $K=1$ and $\mathcal{Q}(2) = 0$ if $K \geq 2$ (recall $M \geq 2$). Since r and run agree for $s \in \{n-2, n-1\}$ and all $s \geq n$, it suffices by Lemma 5.3(d) to show, for $0 \leq s \leq n-3$ and $y = n-s-2 \geq 1$,

$$(6.1) \quad (n-s)T(s) - (n-s-2)T(s+2) + 2H(s+1) + \mathbf{1}[y \text{ odd}](G(s) - G(s+2)) = 2(r_s - r_{s+2}).$$

Since $s+1 = h$ iff $y = K$, the first three terms equal

$$\begin{aligned} & (y+2)(\vartheta(y+2) + \vartheta(y+1)) - y(\vartheta(y+1) + \vartheta(y)) - 2\alpha \mathbf{1}[y = K] \\ &= \frac{\mathcal{Q}(y+2) - \mathcal{Q}(y+1)}{y+1} - 2\alpha \mathbf{1}[y = K], \end{aligned}$$

by the definition of ϑ (at $y+2 \geq 3$ and $y+1 \geq 2$). By Lemma 6.2 at $x = y+1$ this is $2\mathbf{1}[y \geq K-1] + 2\mathbf{1}[y \geq K] - 2\mathbf{1}[y \geq M-1] - 2\mathbf{1}[y \geq M] - \alpha \mathbf{1}[y = 2K+1]$. Next, $G(s) - G(s+2) = \alpha \mathbf{1}[s \in \{2h-n-2, 2h-n-1\}]$, and $s = 2h-n-1$ iff $y = 2K+1$ (odd), while $s = 2h-n-2$ iff $y = 2K+2$ (even); so the last term of (6.1) is $\alpha \mathbf{1}[y = 2K+1]$. Hence the left side of (6.1) is $2\mathbf{1}[K-1 \leq y \leq M-2] + 2\mathbf{1}[K \leq y \leq M-1]$. On the right, $r_s - r_{s+2} = \mathbf{1}[\ell \leq s \leq h] + \mathbf{1}[\ell \leq s+1 \leq h]$, and $\ell \leq s \leq h$ iff $K-1 \leq y \leq M-2$, $\ell \leq s+1 \leq h$ iff $K \leq y \leq M-1$.

(b) By Lemma 5.4 with $T - H = \alpha \mathbf{1}[\cdot = h]$,

$$2 \text{row}_{r+1} - 2 \text{row}_r$$

$$= \alpha (\mathbf{1}[r = h-1] - \mathbf{1}[r = n-1-h]) + G(|2r+3-n|) - G(|2r+1-n|).$$

If $\alpha = 0$ this vanishes. Let $2h \geq n$. One checks $|2r + 1 - n| \leq 2h - n - 1$ iff $n - h \leq r \leq h - 1$; hence $\mathbf{G}(|2r + 3 - n|) - \mathbf{G}(|2r + 1 - n|) = \alpha(\mathbf{1}[n - h - 1 \leq r \leq h - 2] - \mathbf{1}[n - h \leq r \leq h - 1]) = \alpha(\mathbf{1}[r = n - h - 1] - \mathbf{1}[r = h - 1])$ (both intervals are empty if $2h = n$). So all rows are equal, and Lemma 5.5 gives their value.

(c) By Lemma 5.2 it suffices to check (5.1) and (5.2). For $0 \leq s < u \leq n - 1$ with $u - s$ odd put $x = n - s, y = n - u$: then $1 \leq y < x \leq n$, $x - y$ is odd, and $\mathbf{T}(s) + \mathbf{H}(u) = \vartheta(x) + \vartheta(y) - \alpha \mathbf{1}[y = K + 1]$. For $s \leq n - 1$ with $n - s$ odd, $x = n - s$ is odd and $\mathbf{T}(s) + \mathbf{G}(s) = \vartheta(x) + \alpha \mathbf{1}[x \geq 2K + 3]$, because $s \leq 2h - n - 1$ iff $x \geq 2n - 2h + 1 = 2K + 3$. $\square$

In Lean: `dipQ_succ`, `dip_run`, `dip_row_succ`, `isIntervalUnif_dip` (`DipoleUnif.lean`). It remains to choose α so that (CELL1) and (CELL2) hold. For this we need a closed form of ϑ , which involves the following special functions.

7. SPECIAL FUNCTIONS AND THE FACTS F1–F6

Definition 7.1. For integers $x \geq 1$, $K \geq 0$ and $k \geq 1$ let

$$\mathcal{B}(x) = \sum_{j \geq 0} \frac{(-1)^j}{x + j}, \quad \sigma(x) = 4\mathcal{B}(x) - \frac{2}{x}, \quad a(K) = 4K^2\mathcal{B}(K + 1) + 1 - 2K,$$

$$\Omega(k) = \sum_{y=k+1}^{2k} (-1)^y \frac{k}{y(y-1)} \in \mathbb{Q}, \quad w(K) = (-1)^K \Omega(K + 1).$$

The series defining $\mathcal{B}$ converges by the alternating series test. $\mathcal{B}$ is Nielsen's beta function at integer arguments: $\mathcal{B}(1) = \ln 2$ and $\mathcal{B}(m + 1) = (-1)^m (\ln 2 - \sum_{i=1}^m (-1)^{i-1}/i)$, which gives the formula for $a(K)$ in the introduction; for instance $a(1) = 3 - 4 \ln 2 \approx 0.2274$, $a(2) = 16 \ln 2 - 11 \approx 0.0904$, $a(3) = 25 - 36 \ln 2 \approx 0.0467$. Also $\Omega(2) = -\frac{1}{6}$, $\Omega(3) = \frac{1}{5}$, $\Omega(4) = -\frac{19}{210}$. These closed forms are not used below. In Lean, $\mathcal{B}$, σ , a , Ω , w are `betaF` (defined as the limit of the partial sums), `sigmaF`, `aF`, `omegaF`, `wF` (`UnifFacts.lean`).

Lemma 7.2 (alternating recursions). *Let $x_0 \in \mathbb{N}$ and $f, c : \{x_0, x_0 + 1, \dots\} \rightarrow \mathbb{R}$ with $f(x) + f(x + 1) = c(x)$ for all $x \geq x_0$, c nonincreasing and $f(x) \rightarrow 0$. Then $0 \leq f(x) \leq c(x)$ for all $x \geq x_0$. If moreover $x \mapsto c(x) - c(x + 1)$ is nonincreasing, then*

$$\frac{1}{2}c(x) \leq f(x) \leq \frac{1}{2}c(x) + \frac{1}{2}(c(x) - c(x + 1)).$$

Proof. $f(x + 2) - f(x) = c(x + 1) - c(x) \leq 0$, so $N \mapsto f(x + 2N)$ is nonincreasing with limit 0; hence $f(x) \geq 0$, and $f(x) = c(x) - f(x + 1) \leq c(x)$. For the second part apply the first to $g(x) = f(x) - f(x + 1)$, which satisfies $g(x) + g(x + 1) = c(x) - c(x + 1)$ and $g \rightarrow 0$: then $0 \leq g(x) \leq c(x) - c(x + 1)$, and $2f(x) = c(x) + g(x)$. $\square$

Lemma 7.3. *For integers $x \geq 1$: $\mathcal{B}(x) + \mathcal{B}(x + 1) = \frac{1}{x}$, $0 \leq \mathcal{B}(x) \leq \frac{1}{x}$, and $\mathcal{B}(x) \rightarrow 0$.*

Proof. The partial sums satisfy $\sum_{j < N+1} \frac{(-1)^j}{x+j} = \frac{1}{x} - \sum_{j < N} \frac{(-1)^j}{x+1+j}$; let $N \rightarrow \infty$. The partial sums of even length are $\leq \mathcal{B}(x)$ and those of odd length are $\geq \mathcal{B}(x)$, which gives $0 \leq \mathcal{B}(x) \leq 1/x$. $\square$

Lemma 7.4 (Lemma B). *Let $u(x) = \frac{1}{2x} + \frac{1}{4x^2} - \frac{1}{8x^4} + \frac{1}{4x^6}$ and $\varepsilon(x) = \frac{17x^4 + 34x^3 + 29x^2 + 12x + 2}{8x^6(x+1)^6}$. For every integer $x \geq 1$,*

$$u(x) - \varepsilon(x) \leq \mathcal{B}(x) \leq u(x).$$

Proof. A direct computation gives $u(x) + u(x + 1) - \frac{1}{x} = \varepsilon(x)$ and $\varepsilon(x) - \varepsilon(x + 1) = \frac{17x^4 + 68x^3 + 100x^2 + 64x + 16}{x^6(x+1)(x+2)^6} > 0$. So $f = u - \mathcal{B}$ satisfies $f(x) + f(x + 1) = \varepsilon(x)$ (Lemma 7.3) with ε nonincreasing, and $f \rightarrow 0$. Lemma 7.2 gives $0 \leq f \leq \varepsilon$. $\square$

Lemma 7.5 (properties of σ). *Let $x, M \geq 1$ and $0 \leq K \leq M$ be integers.*

- (i) $\sigma(x) + \sigma(x+1) = \frac{2}{x(x+1)}$.
- (ii) $\frac{1}{x(x+1)} \leq \sigma(x) \leq \frac{x+4}{x(x+1)(x+2)}$; in particular $\sigma(x) > 0$, $\sigma(4) \leq \frac{1}{15}$ and $\sigma(5) \leq \frac{3}{70}$.
- (iii) $\sigma(x+1) < \sigma(x)$.
- (iv) $a(K) = \frac{2}{K+1} - 1 + K^2\sigma(K+1)$, and $a(M) = 1 - M^2\sigma(M)$.
- (v) $M^2\sigma(M+1) \leq 1 - \frac{1}{M+1}$.
- (vi) $(M^2 - K^2)\sigma(x) \leq 1 - \frac{1}{M+1}$ for $x \geq M+1$.
- (vii) $(M^2 - K^2)\sigma(M+1) + a(M) \leq 1 - K^2\sigma(M+1)$.
- (viii) $1 - K^2\sigma(x) \geq \frac{2}{K+1} - a(K)$ for $x \geq K+1$.

Proof. (i) follows from $\mathcal{B}(x) + \mathcal{B}(x+1) = 1/x$. (ii) Apply the second part of Lemma 7.2 to $f = \sigma$, $c(x) = \frac{2}{x(x+1)}$, for which $c(x) - c(x+1) = \frac{4}{x(x+1)(x+2)}$ is nonincreasing, and $\sigma \rightarrow 0$. (iii) By (ii) at $x+1$ and since $x(x+5) < (x+2)(x+3)$, $\sigma(x+1) \leq \frac{x+5}{(x+1)(x+2)(x+3)} < \frac{1}{x(x+1)}$; by (i), $\sigma(x) = \frac{2}{x(x+1)} - \sigma(x+1) > \frac{1}{x(x+1)} > \sigma(x+1)$. (iv) The first identity is the definition rewritten with $\sigma(K+1) = 4\mathcal{B}(K+1) - \frac{2}{K+1}$; the second follows from the first (at $K = M$) and (i) at $x = M$. (v) By (ii), $M^2\sigma(M+1) \leq \frac{M^2(M+5)}{(M+1)(M+2)(M+3)} \leq \frac{M}{M+1}$, since $M(M+5) \leq (M+2)(M+3)$. (vi) By (iii), $0 < \sigma(x) \leq \sigma(M+1)$, and $0 \leq M^2 - K^2 \leq M^2$; use (v). (vii) By (iv), the left side is $2M^2\sigma(M+1) - K^2\sigma(M+1) + \frac{2}{M+1} - 1$; use (v). (viii) By (iii) and (iv), $1 - K^2\sigma(x) \geq 1 - K^2\sigma(K+1) = \frac{2}{K+1} - a(K)$. $\square$

Lemma 7.6. For $k \geq 1$, $\Omega(k) = (-1)^k(2k\mathcal{B}(k+1) - 1) + 2k\mathcal{B}(2k) - \frac{1}{2}$. Consequently, for $K \geq 0$,

$$w(K) = \begin{cases} \frac{3}{2} - 2(K+1)(\mathcal{B}(K+2) + \mathcal{B}(2K+2)), & K \text{ odd}, \\ \frac{1}{2} - 2(K+1)\mathcal{B}(K+2) + 2(K+1)\mathcal{B}(2K+2), & K \text{ even}. \end{cases}$$

Proof. Write $\frac{k}{y(y-1)} = k(\frac{1}{y-1} - \frac{1}{y})$, so $\Omega(k) = -k(\sum_{z=k}^{2k-1} \frac{(-1)^z}{z} + \sum_{y=k+1}^{2k} \frac{(-1)^y}{y})$. By Lemma 7.3, $\sum_{z=a}^{a+N-1} \frac{(-1)^z}{z} = (-1)^a\mathcal{B}(a) - (-1)^{a+N}\mathcal{B}(a+N)$. Hence $\Omega(k) = -k((-1)^k(\mathcal{B}(k) - \mathcal{B}(k+1)) - \mathcal{B}(2k) + \mathcal{B}(2k+1))$, and $\mathcal{B}(k) = \frac{1}{k} - \mathcal{B}(k+1)$, $\mathcal{B}(2k+1) = \frac{1}{2k} - \mathcal{B}(2k)$ give the formula. For $w(K)$ take $k = K+1$ and multiply by $(-1)^K$. $\square$

Proposition 7.7 (facts F1–F6). F1 $a(K) > 0$ for all $K \geq 0$.

F2 $a(K+1) < a(K)$ for all $K \geq 0$.

F3 $a(K) + a(K+1) \leq \frac{1}{2(K+1)}$ for all $K \geq 2$.

F4 $w(K) > 0$ and $(K+1)(a(K) + a(K+1)) \leq 4w(K)$ for all $K \geq 2$.

F5 $a(1) - \frac{2}{21} - a(3) > 0$, $a(1) - \frac{2}{21} + a(2) \leq \frac{2}{7}$, and $a(1) < \frac{5}{21}$.

F6 $a(1) < \frac{1}{4}$ and $a(3) < \frac{1}{20}$.

Proof. Put $a^+(K) = 4K^2u(K+1) + 1 - 2K$ and $a^-(K) = 4K^2(u(K+1) - \varepsilon(K+1)) + 1 - 2K$. By Lemma 7.4 at $x = K+1$, $a^-(K) \leq a(K) \leq a^+(K)$ for all $K \geq 0$.

F1: $a(K) \geq a^-(K)$, which is a quotient of polynomials with positive coefficients by (A.1).

F2: $a(K) - a(K+1) \geq a^-(K) - a^+(K+1) > 0$ by (A.2).

F3: for $K = k+2$, $\frac{1}{2(K+1)} - a(K) - a(K+1) \geq \frac{1}{2(K+1)} - a^+(K) - a^+(K+1) \geq 0$ by (A.3).

F4: let $K = k+2 \geq 2$. If K is odd, Lemma 7.6 and the upper bounds of Lemma 7.4 at $K+2$ and $2K+2$ give $w(K) \geq w_{\text{odd}}^-(K) := \frac{3}{2} - 2(K+1)(u(K+2) + u(2K+2))$, which is positive by (A.4); and by (A.5),

$$(K+1)(a(K) + a(K+1)) \leq (K+1)(a^+(K) + a^+(K+1)) \leq 4w_{\text{odd}}^-(K) \leq 4w(K).$$

If K is even, the upper bound at $K+2$ and the lower bound at $2K+2$ give $w(K) \geq w_{\text{even}}^-(K) := \frac{1}{2} - 2(K+1)u(K+2) + 2(K+1)(u(2K+2) - \varepsilon(2K+2))$, which is positive by (A.6), and (A.7) gives the second inequality in the same way.

F5, F6: Lemma 7.4 at $x = 2, 3, 4$ gives exact rational enclosures. With $u(2) = \frac{79}{256}$, $u(3) = \frac{1127}{5832}$, $u(4) = \frac{2297}{16384}$ and the values of ε ,

$$a(1) \in \left[\frac{15}{64} - \frac{343}{46656}, \frac{15}{64} \right], \quad a(2) \in \left[\frac{67}{729} - \frac{1297}{746496}, \frac{67}{729} \right],$$

$$a(3) \in \left[\frac{193}{4096} - \frac{31689}{64000000}, \frac{193}{4096} \right],$$

that is $0.22702 \leq a(1) \leq 0.23438$, $0.09016 \leq a(2) \leq 0.09191$ and $0.04662 \leq a(3) \leq 0.04712$. Hence $a(1) - \frac{2}{21} - a(3) \geq 0.0846$, $a(1) - \frac{2}{21} + a(2) \leq 0.2311 < \frac{2}{7}$, $a(1) \leq \frac{15}{64} < \frac{5}{21}$, $a(1) < \frac{1}{4}$ and $a(3) \leq \frac{193}{4096} < \frac{1}{20}$. $\square$

The polynomial identities (A.1)–(A.7) are listed in Appendix A; they are identities of rational functions, which can be checked by clearing denominators and expanding. In Lean they are proved by `field_simp`; `ring (poly_F1 to poly_F4d)`, and F1–F6 are `factF1` to `factF6` (`UnifFacts.lean`); the Lean proof of F5, F6 uses the slightly weaker decimal enclosures $0.227 < a(1) < 0.2344$, $0.0901 < a(2) < 0.092$, $0.0466 < a(3) < 0.0472$.

Corollary 7.8. (a) $a(L) \leq a(K)$ for $L \geq K \geq 0$. (b) $a(K) < \frac{1}{2(K+1)}$ for $K \geq 1$.

Proof. (a) is F2. (b) For $K = 1$ this is F6; for $K \geq 2$ it follows from F3 and $a(K+1) > 0$ (F1). $\square$

8. CLOSED FORM OF ϑ

Throughout this section we are in the setting of Definition 6.1 with any $\alpha \in \mathbb{R}$. Let $\Gamma = M^2 - K^2$ and, for integers $x, y \geq 1$,

$$\Phi(x) = \begin{cases} 1 - K^2\sigma(x), & x \leq M, \\ \Gamma\sigma(x), & x \geq M+1, \end{cases} \quad b(y) = \begin{cases} \frac{2(K+1)\alpha}{y(y-1)}, & K+2 \leq y \leq 2K+2, \\ 0, & \text{otherwise,} \end{cases}$$

and $\psi(x) = \vartheta(x) - \Phi(x)$ (the “alternating part”).

Lemma 8.1. (a) $\vartheta(x) = 0$ for $0 \leq x \leq K$. (b) $\vartheta(K+1) = \frac{2}{K+1}$ and $\psi(K+1) = a(K)$. (c) For $x \geq K+1$, $\psi(x+1) = -\psi(x) + b(x+1) - \mathbf{1}[x = M]a(M)$.

Proof. (a) $\mathcal{Q}(x) = 0$ for $x \leq K$, since $x < K+1 < M+1$ and $x < K+2$; so $\vartheta(x) = -\vartheta(x-1) = 0$ for $2 \leq x \leq K$ by induction. (b) $\mathcal{Q}(K+1) = q_K(K+1) = 2(K+1)K - 2K^2 = 2K$, so $\vartheta(K+1) = \frac{2K}{(K+1)K} - \vartheta(K) = \frac{2}{K+1}$; then $\psi(K+1) = \frac{2}{K+1} - 1 + K^2\sigma(K+1) = a(K)$ by Lemma 7.5(iv). (c) For $x \geq K+1$, $\vartheta(x+1) + \vartheta(x) = \mathcal{Q}(x+1)/(x(x+1))$ with

$$\mathcal{Q}(x+1) = \begin{cases} 2x(x+1) - 2K^2 + b(x+1)x(x+1), & x+1 \leq M, \\ 2M^2 - 2K^2 + b(x+1)x(x+1), & x+1 \geq M+1. \end{cases}$$

If $x+1 \leq M$, Lemma 7.5(i) gives $\Phi(x+1) + \Phi(x) = 2 - \frac{2K^2}{x(x+1)}$, so $\psi(x+1) + \psi(x) = b(x+1)$. If $x \geq M+1$, $\Phi(x+1) + \Phi(x) = \frac{2\Gamma}{x(x+1)}$ and again $\psi(x+1) + \psi(x) = b(x+1)$. If $x = M$, $\Phi(M+1) + \Phi(M) = \Gamma\sigma(M+1) + 1 - K^2\sigma(M) = \frac{2\Gamma}{M(M+1)} + 1 - M^2\sigma(M) = \frac{2\Gamma}{M(M+1)} + a(M)$ by Lemma 7.5(i),(iv), so $\psi(M+1) + \psi(M) = b(M+1) - a(M)$. $\square$

Unwinding the recursion, for $x \geq K+1$,

$$(8.1) \quad \psi(x) = (-1)^x \left((-1)^{K+1}a(K) + \sum_{y=K+2}^x (-1)^y b(y) + \mathbf{1}[x \geq M+1](-1)^M a(M) \right).$$

Remark 8.2 (the parity mode and the dipole). For $x \geq \max(M+1, 2K+2)$, (8.1) and $\sum_{y=K+2}^{2K+2} (-1)^y b(y) = 2\alpha\Omega(K+1)$ give $\vartheta(x) = \Gamma\sigma(x) + (-1)^x (A_{\text{ren}} + 2\alpha\Omega(K+1))$ with $A_{\text{ren}} = (-1)^{K+1}a(K) + (-1)^Ma(M)$, which is nonzero because $a(K) > a(M) > 0$. Without a dipole ($\alpha = 0$) the smooth part $\Gamma\sigma(x) \approx \Gamma/x^2$ eventually becomes smaller than the parity amplitude $|A_{\text{ren}}|$, and cells can become negative when n is large compared with K : for example, for $(n, \ell, h) = (12, 10, 10)$ the matrix $Y^0 = \mathcal{R}_{12}^{10} - \mathcal{R}_{12}^9$ has minimal entry $-677/4620$, attained at the central cells $(5, 6)$ and $(6, 5)$. (That $Y^0 = \mathcal{R}_n^h - \mathcal{R}_n^{\ell-1}$ in general is not needed; it was checked for $n \leq 20$, Appendix B.) The dipole amplitude

$$(8.2) \quad \alpha^* = \alpha^*(K, M) = -\frac{A_{\text{ren}}}{2\Omega(K+1)} = \frac{a(K) - (-1)^{K+M}a(M)}{2w(K)}$$

cancels this parity mode exactly. (In Lean: `alphaStar`.)

In Lean, Lemma 8.1 is `dipTau_low`, `dipTau_base`, `dipS_base` and `dipS_succ` (`UnifAllN.lean`).

9. THE CASE ANALYSIS

Throughout, $1 \leq \ell \leq h \leq n-2$, $K = n-1-h$, $M = n-\ell$, and ϑ, Φ, ψ, b are as in Sections 6–8. Every pair $(n, [\ell, h])$ falls into exactly one of the cases (C) $n \leq 2K+1$; (D) $n \geq 2K+2$ and $K=1$; (E) $n \geq 2K+2$ and $K \geq 2$. We repeatedly use that $(-1)^{x+c} = -(-1)^{y+c}$ when $x-y$ is odd, and Corollary 7.8: for $M \geq K+1$, $0 < a(M) \leq a(K+1) < a(K)$.

9.1. Case (C): no dipole.

Proposition 9.1. *If $n \leq 2K+1$, then Y^0 satisfies (CELL1) and (CELL2); hence it is an interval uniformisation of $[\ell, h]$.*

Proof. If $K=1$ then $n \leq 3$, which forces $(n, \ell, h) = (3, 1, 1)$ and $M=2$. Here $\vartheta(0), \dots, \vartheta(3) = 0, 0, 1, 0$ ($\mathcal{Q}(2) = 2$, $\mathcal{Q}(3) = q_1(3) - q_2(3) = 10 - 4 = 6$), the pairs in (CELL1) are $(x, y) = (2, 1), (3, 2)$ with sums 1, 1, and $\vartheta(1) = \vartheta(3) = 0$.

Let $K \geq 2$ and $\alpha = 0$, so $b \equiv 0$. By Lemma 8.1 and induction, for $x \geq K+1$,

$$\vartheta(x) = \Phi(x) + (-1)^{x+K+1}a(K) + \mathbf{1}[x \geq M+1](-1)^{x+M}a(M).$$

Bounds for single $x \leq n$. If $x \leq K$, $\vartheta(x) = 0$. If $K+1 \leq x \leq M$, by Lemma 7.5(viii) and Corollary 7.8(b), $\vartheta(x) \geq \frac{2}{K+1} - 2a(K) > \frac{2}{K+1} - \frac{1}{K+1} > 0$. If $M+1 \leq x \leq n \leq 2K+1$, then $\Gamma \geq (K+1)^2 - K^2 = 2K+1$ and, by Lemma 7.5(ii),(iii), $\sigma(x) \geq \sigma(2K+1) \geq \frac{1}{(2K+1)(2K+2)}$, so $\vartheta(x) \geq \Gamma\sigma(x) - a(K) - a(K+1) \geq \frac{1}{2(K+1)} - a(K) - a(K+1) \geq 0$ by F3. For the upper bound, $\Phi(x) \leq 1$ (for $x \leq M$ since $\sigma > 0$; for $x > M$ by Lemma 7.5(vi)), so $\vartheta(x) \leq 1 + a(K) + a(K+1) \leq 1 + \frac{1}{2(K+1)} < 2$ by F3.

Pairs. Let $K+1 \leq y < x \leq n$ with $x-y$ odd. The $a(K)$ terms cancel. If $x \leq M$: $\vartheta(x) + \vartheta(y) = 2 - K^2(\sigma(x) + \sigma(y)) \leq 2$. If $y \leq M < x$: using $\Gamma\sigma(x) \leq \Gamma\sigma(M+1)$ and Lemma 7.5(vii), $\vartheta(x) + \vartheta(y) \leq \Gamma\sigma(M+1) + a(M) + 1 - K^2\sigma(y) \leq 2 - K^2\sigma(M+1) - K^2\sigma(y) \leq 2$. If $M < y$: the $a(M)$ terms cancel too, and $\vartheta(x) + \vartheta(y) = \Gamma(\sigma(x) + \sigma(y)) \leq 2 - \frac{2}{M+1}$ by Lemma 7.5(vi).

Conclusion. In (CELL1) (with $\alpha = 0$) both terms are ≥ 0 ; if $y \leq K$ then $\vartheta(y) = 0$ and $\vartheta(x) \leq 2$, otherwise the pair bound applies. (CELL2) is $0 \leq \vartheta(x) \leq 2$. $\square$

In Lean: `dipS_C`, `caseC_bounds`, `caseC_cells`, `caseC_K1_cells`.

9.2. Case (E): one dipole of amplitude α^* .

Lemma 9.2. *Let $\alpha \geq 0$ and $\text{tail}(x) = \sum_{j \geq 0} (-1)^j b(x+1+j)$ (a finite sum). Then $\text{tail}(x) = b(x+1) - \text{tail}(x+1)$, $\text{tail}(x) = 0$ for $x \geq 2K+2$, $\text{tail}(K+1) = 2\alpha w(K)$, and $0 \leq \text{tail}(x) \leq b(x+1)$ for $x \geq K+1$.*

Proof. The first two statements are immediate. With $y = K+2+j$, $\text{tail}(K+1) = \sum_{j=0}^K (-1)^j \frac{2(K+1)\alpha}{(K+2+j)(K+1+j)}$, while $w(K) = (-1)^K \sum_{y=K+2}^{2K+2} (-1)^y \frac{K+1}{y(y-1)} = \sum_{j=0}^K (-1)^j \frac{K+1}{(K+2+j)(K+1+j)}$. For $x \geq K+1$ the sequence $b(x+1), b(x+2), \dots$ is nonnegative and nonincreasing, so its alternating sum lies in $[0, b(x+1)]$. $\square$

Lemma 9.3. *If $K \geq 2$ and $M \geq K+1$, then $0 < \alpha^* \leq \frac{2}{K+1}$.*

Proof. By F4, $w(K) > 0$. The numerator of (8.2) is $\geq a(K) - a(M) > 0$ and $\leq a(K) + a(M) \leq a(K) + a(K+1) \leq \frac{4w(K)}{K+1}$ by F4. $\square$

Proposition 9.4. *If $K \geq 2$ and $n \geq 2K+2$, then Y^{α^*} satisfies (CELL1) and (CELL2); hence it is an interval uniformisation of $[\ell, h]$.*

Proof. Write $\alpha = \alpha^*$. By Proposition 6.3(b) the condition $2h \geq n$ is needed for the rows; it is equivalent to $n \geq 2K+2$. We show the cell inequalities for all $x, y \geq 1$, not only $x \leq n$.

Closed form. For $x \geq K+1$,

$$(9.1) \quad \psi(x) = \text{tail}(x) - \mathbf{1}[x \leq M](-1)^{x+M}a(M).$$

For $x = K+1$: by Lemma 9.2 and (8.2), the right side is $2\alpha w(K) + (-1)^{K+M}a(M) = a(K) = \psi(K+1)$. If (9.1) holds at $x \geq K+1$, then by Lemma 8.1(c) and Lemma 9.2, $\psi(x+1) = \text{tail}(x+1) + \mathbf{1}[x \leq M](-1)^{x+M}a(M) - \mathbf{1}[x = M]a(M)$, which is (9.1) at $x+1$ (for $x < M$ directly; for $x = M$ the two $a(M)$ terms cancel; for $x > M$ both vanish).

Auxiliary bounds. By Lemma 9.3, $(K+1)\alpha \leq 2$, so for $z \geq K+1$, $\text{tail}(z) \leq b(z+1) \leq \frac{4}{z(z+1)} \leq K^2\sigma(z)$ (Lemma 7.5(ii) and $K^2 \geq 4$) and $\text{tail}(z) \leq \frac{4}{12} = \frac{1}{3}$ (as $z \geq 3$). Also $a(M) \leq a(1) < \frac{1}{4}$ and $a(M) \leq a(K+1) < a(K) < \frac{1}{2(K+1)}$.

Lower bound $\vartheta \geq 0$. For $x \leq K$, $\vartheta(x) = 0$. For $K+1 \leq x \leq M$, by Lemma 7.5(viii), $\vartheta(x) \geq \frac{2}{K+1} - a(K) + 0 - a(M) > \frac{2}{K+1} - 2a(K) > 0$. For $x \geq M+1$, $\vartheta(x) = \Gamma\sigma(x) + \text{tail}(x) \geq 0$.

Upper bounds. $\Phi(x) \leq 1$ for $x \geq K+1$, so $\vartheta(x) \leq 1 + \frac{1}{3} + \frac{1}{4} < 2$ for all x . For $x \geq 2K+3$, $\text{tail}(x) = 0$ and $\alpha \leq \frac{2}{K+1} \leq \frac{2}{3}$, so $\vartheta(x) + \alpha \leq 1 + \frac{1}{4} + \frac{2}{3} < 2$.

Pairs. Let $K+1 \leq y < x$ with $x-y$ odd; we show $\vartheta(x) + \vartheta(y) \leq 2$. If $x \leq M$, the $a(M)$ terms cancel and $\vartheta(x) + \vartheta(y) = 2 - K^2\sigma(x) + \text{tail}(x) - K^2\sigma(y) + \text{tail}(y) \leq 2$. If $y \leq M < x$, then $\text{tail}(x) \leq \frac{4}{(M+1)(M+2)} \leq K^2\sigma(M+1)$ and $\text{tail}(y) \leq K^2\sigma(y)$, so by Lemma 7.5(vii),

$$\begin{aligned} \vartheta(x) + \vartheta(y) &\leq (\Gamma\sigma(M+1) + a(M)) + \text{tail}(x) + 1 - K^2\sigma(y) + \text{tail}(y) \\ &\leq (1 - K^2\sigma(M+1)) + K^2\sigma(M+1) + 1 = 2. \end{aligned}$$

If $M < y$, then $\vartheta(x) + \vartheta(y) \leq 2(1 - \frac{1}{M+1}) + \frac{4}{x(x+1)} + \frac{4}{y(y+1)} \leq 2$, since $x, y \geq M+1 \geq 4$ gives $\frac{4}{z(z+1)} \leq \frac{1}{M+1}$ for $z \in \{x, y\}$.

Conclusion. (CELL1): if $y = K+1$, then $\vartheta(y) - \alpha = \frac{2}{K+1} - \alpha \geq 0$ gives the lower bound, and $\vartheta(x) + \vartheta(K+1) - \alpha \leq \vartheta(x) + \vartheta(K+1) \leq 2$ by the pair bound. If $y \neq K+1$, the lower bound is $\vartheta \geq 0$; the upper bound is the single bound if $y \leq K$ and the pair bound if $y \geq K+2$. (CELL2): for $x \geq 2K+3$ use $\vartheta(x) + \alpha \leq 2$, otherwise $0 \leq \vartheta(x) < 2$. $\square$

In Lean: `alphaStar_bounds`, `dipTail_base`, `dipTail_bounds`, `dipS_E`, `caseE_cells_gen`, `caseE_cells`.

9.3. Case (D): $K = 1$ and one dipole of amplitude $2/7$.

Proposition 9.5. *If $K = 1$ (i.e. $h = n - 2$) and $n \geq 4$, then $Y^{2/7}$ satisfies (CELL1) and (CELL2); hence it is an interval uniformisation of $[\ell, h]$.*

Proof. Let $\alpha = \frac{2}{7}$; then $2h = 2n - 4 \geq n$. Here $2 \leq M \leq n - 1$, $\Gamma = M^2 - 1$, $b(3) = \frac{4}{21}$, $b(4) = \frac{2}{21}$ and $b(y) = 0$ for $y \geq 5$. Put $e_P = a(1) - \frac{2}{21}$ and $A = e_P + (-1)^M a(M)$.

Values. $\vartheta(1) = 0$ and $\vartheta(2) = 1$ (Lemma 8.1). By Lemma 8.1, $\psi(2) = a(1)$, $\psi(3) = -a(1) + \frac{4}{21} - 1[M = 2]a(2)$, and for $x \geq 4$

$$\psi(x) = (-1)^x (e_P + \mathbf{1}[x \geq M + 1](-1)^M a(M))$$

(check $x = 4$ for $M = 2$, $M = 3$, $M \geq 4$ separately, then induct with $b(x + 1) = 0$). Since $a(1) = \sigma(2)$, $a(2) = 4\sigma(3) - \frac{1}{3}$ and $\sigma(2) + \sigma(3) = \frac{1}{3}$ (Lemma 7.5(i),(iv)), we get $\vartheta(3) = \frac{6}{7}$ if $M \geq 3$ and $\vartheta(3) = \frac{4}{21}$ if $M = 2$. For $x \geq 4$: $\vartheta(x) = 1 - \sigma(x) + (-1)^x e_P$ if $x \leq M$, and $\vartheta(x) = \Gamma\sigma(x) + (-1)^x A$ if $x \geq M + 1$.

Constants. By F5 and F1, $0 < a(1) - \frac{2}{21} - a(3) < e_P < \frac{5}{21} - \frac{2}{21} = \frac{1}{7}$. If M is even, $A = e_P + a(M) \in (0, e_P + a(2)] \subseteq (0, \frac{2}{7}]$ by F2 and F5; if M is odd, then $M \geq 3$ and $A = e_P - a(M) \geq e_P - a(3) > 0$, $A < e_P < \frac{2}{7}$. By Lemma 7.5(ii),(iii), $\sigma(x) \leq \frac{1}{15}$ for $x \geq 4$ and $\sigma(x) \leq \frac{3}{70}$ for $x \geq 5$. For $x \geq M + 1$: $0 \leq \Gamma\sigma(x) \leq 1 - \frac{1}{M+1}$ and $\Gamma\sigma(x) + a(M) \leq \Gamma\sigma(M + 1) + a(M) \leq 1 - \sigma(M + 1) < 1$ (Lemma 7.5(vi),(vii)).

Single $x \geq 4$. Even x : $0 < \vartheta(x) \leq 1 + e_P$ (for $x \leq M$, $\vartheta(x) \geq \frac{14}{15}$; for $x > M$, $\vartheta(x) = \Gamma\sigma(x) + A > 0$ and $\Gamma\sigma(x) + A \leq \Gamma\sigma(x) + a(M) + e_P < 1 + e_P$). Odd x : $-\frac{2}{7} \leq \vartheta(x) \leq 1$ (for $x \leq M$, $x \geq 5$ and $\vartheta(x) = 1 - \sigma(x) - e_P \in [1 - \frac{3}{70} - \frac{1}{7}, 1]$; for $x > M$, $\vartheta(x) = \Gamma\sigma(x) - A \in [-A, 1 - A]$).

Pairs $4 \leq y < x$, $x - y$ odd. If $x \leq M$: $\vartheta(x) + \vartheta(y) = 2 - \sigma(x) - \sigma(y) \in [0, 2]$. If $y \leq M < x$: $\vartheta(x) + \vartheta(y) = \Gamma\sigma(x) + 1 - \sigma(y) - (-1)^{y+M} a(M)$, which is $\leq \Gamma\sigma(M + 1) + a(M) + 1 - \sigma(y) \leq 2$ and $\geq 1 - \frac{1}{15} - a(2) > 0$. If $M < y$: $\vartheta(x) + \vartheta(y) = \Gamma(\sigma(x) + \sigma(y)) \in [0, 2]$.

Conclusion. (CELL1) with $y \geq 4$ is the pair bound ($y \neq K + 1 = 2$). For $y = 1$ (x even): $\vartheta(x) \in \{1\} \cup [0, 1 + e_P]$. For $y = 2$ (x odd, $\vartheta(2) - \alpha = \frac{5}{7}$): $\vartheta(3) + \frac{5}{7} \in \{\frac{19}{21}, \frac{11}{7}\}$, and $\vartheta(x) + \frac{5}{7} \in [\frac{3}{7}, \frac{12}{7}]$ for $x \geq 5$. For $y = 3$ ($x \geq 4$ even): $0 \leq \vartheta(x) + \vartheta(3) \leq 1 + e_P + \frac{6}{7} < 2$. (CELL2): $\vartheta(1) = 0$, $\vartheta(3) \in [0, 2]$, and for odd $x \geq 5$ (where $x \geq 2K + 3$), $\vartheta(x) + \frac{2}{7} \in [0, \frac{9}{7}]$. $\square$

In Lean: `dipB_D3`, `dipB_D4`, `dipS_D`, `tauD_three`, `caseD_cells`.

9.4. Proof of Theorem 3. Let $n \geq 1$ and $0 \leq \ell \leq h \leq n - 1$. Case (A), $h = n - 1$: take J_n if $\ell = 0$ and $J_n - \mathcal{R}_n^{\ell-1}$ if $\ell \geq 1$ (Lemma 4.6). Case (B), $\ell = 0$ and $h \leq n - 2$: take the renewal square $\mathcal{R}_n^h$ (Proposition 4.5). Otherwise $1 \leq \ell \leq h \leq n - 2$, and Propositions 9.1, 9.5 and 9.4 cover the three remaining cases (C), (D), (E). $\square$

In Lean this is `intervalUnifAll_all` (`UnifAllN.lean`).

10. FORMAL VERIFICATION

10.1. What is formalised. Theorem 2 is formalised, with its complete proof, in Lean 4 [9] using Mathlib [19]. The development is the library `STUProof` in the directory `lean/` of the repository [1] (<https://github.com/anshM123/IQOQI-Conjecture>). The final theorem is `STUPProof.stu_exists_unconditional` in `KirwanFreeFinal.lean`; its statement is displayed in Remark 2.1. All lemmas, propositions and theorems of Sections 3–9 are formalised (Lemma 3.12 in the inequality form stated after it), including Lemma 3.4 (Horn’s theorem), the analytic estimates of Section 7 for all K (not by finite computation), and the polynomial identities of Appendix A. The function $\mathcal{B}$ is defined as the limit of its partial sums, whose existence comes from Mathlib’s alternating series test.

The reduction of Theorem 1 to Theorem 2 is `stu_exists_hermitian` (`GeneralH.lean`). Not formalised are the physical discussion of Section 1, the closed forms involving $\ln 2$ mentioned after Definition 7.1 (they are not used), the remarks of Section 11 (except for the ingredients of the Kirwan route, see Section 11.2), and the side remarks marked as checked by computer in Appendix B.

10.2. Versions and checking. The toolchain is Lean v4.33.1 (file `lean-toolchain`), and Mathlib is pinned to commit `0df444a360eaa60ab8c11dca51a86af692955474` (file `lakefile.toml`). The script `check.sh` elaborates all files sequentially (to bound memory), then runs `Axioms.lean`, which prints the axioms used by 67 main declarations, and finally searches the sources for the keywords `sorry`, `admit`, `native_decide` and for `axiom` declarations. A complete run (log `logs/lean_check_final_unconditional.log` in [1]) finished with exit code 0 after 1573 seconds (about 26 minutes) and a peak memory use of roughly 7 GB. For every one of the printed declarations, including `stu_exists_unconditional`, the output has the form

'STUProof.stu_exists_unconditional' depends on axioms:

[`propext`, `Classical.choice`, `Quot.sound`]

that is, only Lean's three standard axioms (propositional extensionality, choice, and quotient soundness) are used, and the keyword search reports no occurrence. In particular no `sorry`, no added axiom and no `native_decide` (which would trust compiled code) is used.

10.3. Trusted base. Besides the Lean kernel, a reader who wants to trust the formal result must check that the statement means what it should. It uses Mathlib's real exponential, finite sums, matrices, the Kronecker product and the unitary group, and four definitions of the project, displayed in Remark 2.1: `gibbs`, `gibbsState`, `partialTraceA` and `partialTraceB`. No other definition of the project enters the statement.

10.4. Correspondence. Table 2 lists the main results of this paper and their Lean counterparts (all in the namespace `STUProof`).

10.5. Other content of the repository. The repository also contains formalised results that are not needed for Theorem 2: the Kirwan route of Section 11.2, with Kirwan's theorem as an explicit hypothesis; unconditional proofs for $d \leq 20$ and $d \leq 40$ from kernel-checked certificates of "hook decompositions" (a stronger combinatorial statement that implies $\text{UNIF}(n)$); and an alternative explicit "cross" construction for interval uniformisations. These are superseded by Theorem 2.

11. REMARKS

11.1. The strong majorization form is false for $d \geq 4$. For $d = 3$, every $q \prec p$ is symmetrically reachable from $p \otimes p$ [2, Thm. 1]. This fails in every higher dimension, so the thermal structure of the target (Lemma 3.7) cannot be dispensed with.

Proposition 11.1. *Let $d \geq 4$, $\delta = \frac{1}{5d}$, $\eta = 2\delta^2$, and*

$$p = (1 - \delta - (d-2)\eta, \delta, \eta, \dots, \eta), \quad s = 1 - p_0 = \delta + (d-2)\eta,$$

$$q = \left(p_0, \frac{s}{d-1}, \dots, \frac{s}{d-1}\right).$$

Then p is strictly positive and nonincreasing (so it is the Gibbs vector of $H = \text{diag}(-\ln p_i)$ at $\beta = 1$), $q \prec p$, but no state ρ in the unitary orbit of $\text{diag}(p \otimes p)$ has both marginals with spectrum q . For $d = 4$: $p = (0.94, 0.05, 0.005, 0.005)$ and $q = (0.94, 0.02, 0.02, 0.02)$.

Paper	Lean declaration(s)	File
Theorem 2	<code>stu_exists_unconditional</code>	<code>KirwanFreeFinal</code>
Proposition 4.4	<code>stu_exists_of_intervalUnif</code>	<code>KirwanFreeU</code>
Theorem 3	<code>intervalUnifAll_all</code>	<code>UnifAllN</code>
Eq. (2.1)	<code>partialTraceB_conj_kronecker,</code> <code>partialTraceA_conj_kronecker</code>	<code>STU</code>
Lemma 3.2	<code>SMaj,</code> <code>SMaj.add,</code> <code>SMaj.smul,</code> <code>SMaj.majOn,</code> <code>smaj_self,</code> <code>smaj_zeroOne</code>	<code>KirwanFreeCore</code>
Lemma 3.4	<code>schurHorn</code>	<code>SchurHorn</code>
Proposition 3.5	<code>ReachTri.orthogonal,</code> <code>tri_reachable,</code> <code>lc_reachable</code>	<code>KirwanFreeCore</code>
Lemma 3.6	<code>ReachTri.convex</code>	<code>KirwanFreeCore</code>
Lemma 3.7, Corollary 3.8	<code>lemmaT,</code> <code>thermal_clip_weights</code>	<code>KirwanFreeCore</code>
Lemmas 3.9, 3.10	<code>clipZ_eq,</code> <code>one_sub_clipZ,</code> <code>wk_cell_identity</code>	<code>KirwanFree</code>
Lemma 3.11	<code>crossInd_run,</code> <code>card_filter_Icc_hook,</code> <code>crossInd_sum</code>	<code>KirwanFreeHook</code>
Lemma 3.12 (inequality form)	<code>leftM_nonneg,</code> <code>Ltot_pos,</code> <code>sum_aM</code>	<code>KirwanFree</code>
Definition 4.1, Lemma 4.2	<code>IsIntervalUnif,</code> <code>IntervalUnifAll,</code> <code>IsIntervalUnif.smaj</code>	<code>IntervalUnif</code>
Proposition 4.3	<code>wkU_reach,</code> <code>wkU_smaj,</code> <code>wkU_rows</code>	<code>KirwanFreeU</code>
Proposition 4.5, Lemma 4.6	<code>isIntervalUnif_renU,</code> <code>isIntervalUnif_boxJ,</code> <code>IsIntervalUnif.compl</code>	<code>Renewal</code>
Lemmas 5.2–5.5	<code>thg_cells,</code> <code>thg_run_rec,</code> <code>thg_row_succ,</code> <code>row_const_value</code>	<code>DipoleUnif</code>
Lemma 6.2, Proposition 6.3	<code>dipQ_succ,</code> <code>dip_run,</code> <code>dip_row_succ,</code> <code>isIntervalUnif_dip</code>	<code>DipoleUnif</code>
Lemmas 7.2–7.6	<code>alt_rec_bounds,</code> <code>alt_rec_bounds2,</code> <code>betaF_add,</code> <code>betaF_bounds,</code> <code>lemmaB,</code> <code>sigmaF_bounds,</code> <code>sigmaF_succ_lt,</code> <code>aF_eq_sigma,</code> <code>omegaF_eq,</code> <code>wF_eq</code>	<code>UnifFacts</code>
Lemma 7.5(v)–(viii)	<code>msig_le,</code> <code>gamma_sig_le,</code> <code>gamma_sig_add_a,</code> <code>phi_low_ge</code>	<code>UnifAllN</code>
Proposition 7.7	<code>factF1,</code> ..., <code>factF6</code>	<code>UnifFacts</code>
Lemma 8.1	<code>dipTau_low,</code> <code>dipTau_base,</code> <code>dipS_base,</code> <code>dipS_succ</code>	<code>UnifAllN</code>
Propositions 9.1, 9.4, 9.5	<code>caseC_cells,</code> <code>caseC_K1_cells,</code> <code>caseE_cells,</code> <code>caseD_cells</code>	<code>UnifAllN</code>

TABLE 2. Paper results and Lean declarations (files in `lean/STUProof/`, extension `.lean`).

Proof. Since $\delta \leq \frac{1}{20}$ and $2(d-2)\delta < \frac{2}{5}$, we have $s \leq \frac{7}{5}\delta$ and $p_0 > \frac{1}{2}$, so $p_0 > \delta > \eta > 0$; q is nonincreasing, and $q \prec p$ because q is the image of p under the doubly stochastic matrix $1 \oplus \frac{1}{d-1}J_{d-1}$. The eigenvalues of $\text{diag}(p \otimes p)$, in nonincreasing order $\lambda_1 \geq \lambda_2 \geq \dots$, start with

$$\lambda_1 = p_0^2 > \lambda_2 = \lambda_3 = p_0\delta > \lambda_4 = \dots = \lambda_{2d-1} = p_0\eta,$$

and all others are products $p_i p_j$ with $i, j \geq 1$, hence $\leq \delta^2 < p_0\eta$ (as $p_0 > \frac{1}{2}$). Note $2\lambda_1 + \lambda_2 + \dots + \lambda_{2d-1} = 2p_0(p_0 + \delta + (d-2)\eta) = 2p_0$, and the remaining eigenvalues sum to s^2 .

Suppose ρ is in the orbit and both marginals have spectrum q . Applying a local unitary $V \otimes W$ (which preserves the orbit and the marginal spectra) we may assume that $|0\rangle$ is an eigenvector

of ρ_A and of ρ_B for the eigenvalue p_0 . Let $P_2 = |00\rangle\langle 00|$ and let P_1 be the projection onto $O = \text{span}\{|0j\rangle, |j0\rangle : j \geq 1\}$, of rank $2(d-1)$. With $X = |0\rangle\langle 0| \otimes 1 + 1 \otimes |0\rangle\langle 0| = 2P_2 + P_1$,

$$\begin{aligned} 2p_0 &= \langle 0|\rho_A|0\rangle + \langle 0|\rho_B|0\rangle = \text{Tr}[X\rho] = \text{Tr}[P_2\rho] + \text{Tr}[(P_2 + P_1)\rho] \\ &\leq \lambda_1 + (\lambda_1 + \dots + \lambda_{2d-1}) = 2p_0, \end{aligned}$$

by Ky Fan's maximum principle [3]. So both Ky Fan bounds are equalities. If a rank- r projection P satisfies $\text{Tr}[P\rho] = \lambda_1 + \dots + \lambda_r$ and $\lambda_r > \lambda_{r+1}$, then P is the spectral projection of ρ for its r largest eigenvalues (writing $\text{Tr}[P\rho] = \sum_i \lambda_i \langle v_i|P|v_i\rangle$ with weights in $[0, 1]$ summing to r , equality forces $\langle v_i|P|v_i\rangle = 1$ for $i \leq r$). Because of the gaps $\lambda_1 > \lambda_2$ and $\lambda_{2d-1} > \lambda_{2d}$, ρ is block diagonal, $\rho = \lambda_1 P_2 \oplus \rho_O \oplus \rho_R$ with respect to $\mathbb{C}|00\rangle \oplus O \oplus R$, $R = \text{span}\{|ij\rangle : i, j \geq 1\}$, where ρ_O has eigenvalues $\lambda_2, \dots, \lambda_{2d-1}$ and $\text{Tr} \rho_R = s^2$.

Now compute ρ_A . The partial trace over B of $|0j\rangle\langle j'0|$ vanishes for $j, j' \geq 1$, and $\text{Tr}_B \rho_R$ is supported on $\text{span}\{|j\rangle : j \geq 1\}$. Hence $\rho_A = \langle 0|\rho_A|0\rangle |0\rangle\langle 0| \oplus (Y + Z)$, where Y is the compression of ρ_O to $\text{span}\{|j0\rangle : j \geq 1\} \cong \mathbb{C}^{d-1}$ and $Z = \text{Tr}_B \rho_R \geq 0$, $\text{Tr} Z = s^2$. Since $\langle 0|\rho_A|0\rangle = p_0$ and ρ_A has spectrum q , all $d-1 \geq 3$ eigenvalues of $Y + Z$ equal $s/(d-1)$. But by Weyl's inequality and Cauchy interlacing [3],

$$\begin{aligned} \lambda_3(Y + Z) &\leq \lambda_3(Y) + \lambda_1(Z) \leq \lambda_3(\rho_O) + \text{Tr} Z = p_0\eta + s^2 \\ &< 2\delta^2 + 2\delta^2 = \frac{4\delta}{5d} < \frac{\delta}{d-1} \leq \frac{s}{d-1}, \end{aligned}$$

using $s^2 \leq \frac{49}{25}\delta^2$. This contradiction proves the claim. $\square$

For $d = 4$ the final inequality reads $0.0083 < 0.02$. The inequalities in this proof were also checked in exact arithmetic for $4 \leq d \leq 400$ (Appendix B).

Conversely, the set of symmetric marginal spectra is not contained in $\{q : q \prec p\}$ either. For $d = 3$ and $p = (0.4, 0.35, 0.25)$, mixing the three cells $(1, 1)$, $(0, 2)$, $(2, 0)$ (distinct rows and columns) to the common diagonal value 0.1075 , which is majorized by $(p_1^2, p_0 p_2, p_2 p_0)$, gives (Lemma 3.4 and the argument of Proposition 3.5) a state whose two marginals are both $\text{diag}(0.4075, 0.335, 0.2575)$; its largest entry exceeds p_0 .

11.2. The Kirwan route. We sketch an alternative proof of Theorem 2, which uses Kirwan's convexity theorem and is *not needed* for the results of this paper. For $k \in [d]$ let $v_k = (\frac{P_k}{k+1}, \dots, \frac{P_k}{k+1}, p_{k+1}, \dots, p_{d-1})$ with $P_k = p_0 + \dots + p_k$.

- (1) Every v_k is a symmetric marginal spectrum of $p \otimes p$, via an explicit real orthogonal matrix: the tail levels $j > k$ are relabelled by cyclic shifts of the top block $T = \{0, \dots, k\}$ (chosen by greedy bin packing, so that each shift class carries weight at most $\frac{1}{k+1}$), and on each cyclic diagonal $\{(a, a+s)\} \subseteq T \times T$ a Schur–Horn rotation realises the image of the diagonal under one doubly stochastic circulant matrix. Lean: `vk_reachable, exists_binPacking (Construction.lean)`.
- (2) $\gamma_{\beta'}(E) \in \text{conv}\{v_0, \dots, v_{d-1}\}$ (Lemma 5 of [2]; Lean: `lemma5, Lemma5.lean`).
- (3) By Kirwan's theorem [13], applied to $U(d) \times U(d)$ acting on the unitary orbit of $\text{diag}(p \otimes p)$ with moment map $\rho \mapsto (\text{Tr}_B \rho, \text{Tr}_A \rho)$, the set of pairs of sorted marginal spectra is a convex polytope [14, 6]; its intersection with the diagonal $\{(q, q)\}$ is convex and contains all v_k , hence $\gamma_{\beta'}(E)$.

In Lean, step 3 is the hypothesis `KirwanSymmetricConvexity d`, and `stu_exists` (`STU.lean`) proves Theorem 2 from it. Kirwan's theorem itself is not formalised.

11.3. Further remarks. More general targets. The argument proves more than Theorem 2: every $q \in \text{conv}\{w_0, \dots, w_{d-1}\}$ belongs to $\text{Reach}(p)$ and is therefore a symmetric marginal of $\text{diag}(p \otimes p)$. By Lemma 3.7 this includes $g(p)/\sum_i g(p_i)$ for every concave nondecreasing $g : [0, \infty) \rightarrow [0, \infty)$ with $\sum_i g(p_i) > 0$. This generalisation is not stated separately in Lean, but it follows from the formalised statements `lemmaT`, `wkU_reach`, `ReachTri.convex` and `ReachTri.orthogonal` by the argument of Proposition 4.4.

Explicitness. The configuration F is explicit: its entries are rational functions of p and of the numbers $a(K)$, $w(K)$ (which involve $\ln 2$). The orthogonal matrix is block diagonal over the $2d-1$ diagonals of the array, and on each diagonal it is a product of Givens rotations and permutations obtained constructively from Lemma 3.4 [5].

Numerical confirmation. Independently of the proof, the whole chain was implemented from the formulas of this paper: for 300 random Hermitian H with $d \leq 6$ (including degenerate spectra) the resulting unitaries have both marginals equal to $\tau_{\beta'}(H)$ up to $4 \cdot 10^{-15}$ (Appendix B).

11.4. Outlook. The set $\mathcal{S}(p)$ of symmetric marginal spectra of $\text{diag}(p \otimes p)$ remains to be understood: we know $\text{conv}\{w_k\} \cup \text{conv}\{v_k\} \subseteq \mathcal{S}(p)$ (the second inclusion via Kirwan), that $\mathcal{S}(p) \neq \{q : q \prec p\}$ for $d \geq 4$, and that $\mathcal{S}(p) \not\subseteq \{q : q \prec p\}$. Natural next questions are the analogous problem for more than two copies, quantitative (robust) versions, and whether the fixed diagonal structure used here also settles other quantum marginal problems with diagonal marginals.

APPENDIX A. POLYNOMIAL IDENTITIES FOR F1–F4

With u, ε from Lemma 7.4 and $a^+, a^-, w_{\text{odd}}^-, w_{\text{even}}^-$ from the proof of Proposition 7.7, the following are identities of rational functions (in (A.3), (A.5) and (A.7), $K = k + 2$). All numerator coefficients are positive.

$$(A.1) \quad a^-(K) = \frac{K^5 + 13K^4 + 70K^3 + 194K^2 + 256K + 128}{2(K+1)(K+2)^6},$$

$$(A.2) \quad a^-(K) - a^+(K+1) = \frac{2K^4 + 23K^3 + 101K^2 + 170K + 98}{2(K+1)(K+2)^6},$$

$$(A.3) \quad \frac{1}{2(K+1)} - a^+(K) - a^+(K+1) = \frac{P_3(k)}{2(k+3)^6(k+4)^7},$$

$$(A.4) \quad w_{\text{odd}}^-(K) = \frac{P_{4a}(K)}{128(K+1)^5(K+2)^6},$$

$$(A.5) \quad \frac{4w_{\text{odd}}^-(K)}{K+1} - a^+(K) - a^+(K+1) = \frac{P_{4b}(k)}{32(k+3)^6(k+4)^6},$$

$$(A.6) \quad w_{\text{even}}^-(K) = \frac{P_{4c}(K)}{4(K+2)^6(2K+3)^6},$$

$$(A.7) \quad \frac{4w_{\text{even}}^-(K)}{K+1} - a^+(K) - a^+(K+1) = \frac{P_{4d}(k)}{2(k+3)^6(k+4)^6(2k+7)^6},$$

where

$$\begin{aligned}
P_3(k) &= k^{12} + 41k^{11} + 764k^{10} + 8545k^9 + 63789k^8 + 334153k^7 + 1256064k^6 + 3400638k^5 \\
&\quad + 6544384k^4 + 8654345k^3 + 7354236k^2 + 3503150k + 660280, \\
P_{4a}(K) &= 48K^{10} + 768K^9 + 5442K^8 + 22428K^7 + 59385K^6 + 105468K^5 + 127292K^4 \\
&\quad + 103200K^3 + 53840K^2 + 16320K + 2176, \\
P_{4b}(k) &= 16k^{10} + 544k^9 + 8242k^8 + 73212k^7 + 421633k^6 + 1641880k^5 + 4368048k^4 \\
&\quad + 7817152k^3 + 8974576k^2 + 5940736k + 1710240, \\
P_{4c}(K) &= 160K^{11} + 3200K^{10} + 29012K^9 + 157296K^8 + 566317K^7 + 1420999K^6 + 2535030K^5 \\
&\quad + 3215114K^4 + 2841125K^3 + 1666285K^2 + 583870K + 92626, \\
P_{4d}(k) &= 192k^{16} + 10816k^{15} + 285192k^{14} + 4672168k^{13} + 53227666k^{12} + 447135728k^{11} \\
&\quad + 2864967224k^{10} + 14282437922k^9 + 55985185735k^8 + 173131057544k^7 \\
&\quad + 420984335806k^6 + 796458028404k^5 + 1149337523504k^4 + 1223024950168k^3 \\
&\quad + 905103455591k^2 + 416231111464k + 89613810730.
\end{aligned}$$

These are the identities `poly_F1`, `poly_F2`, `poly_F3`, `poly_F4a`, `poly_F4b`, `poly_F4c`, `poly_F4d` of `UnifFacts.lean`, where they are proved by `field_simp`; `ring`. We also verified them independently with exact computer algebra (Appendix B).

APPENDIX B. INDEPENDENT COMPUTER CHECKS

The proof does not depend on any computation beyond the formal verification. To guard against transcription errors between the Lean development and this paper, every formula stated here was re-implemented from the paper and checked; the scripts and their logs are available with the source files of this paper (directory `checks/`).

- `check_identities.py` (exact computer algebra): the ingredients of Lemma 7.4, the identities of Lemmas 7.5, 7.6 and 6.2, the renewal identities of Proposition 4.5, the Hall identity of Lemma 3.12, and (A.1)–(A.7) with positivity of all coefficients.
- `check_facts_numeric.py` and `check_small_values.py`: F1–F6 and Corollary 7.8 at 60 digits for $K \leq 3000$ (F4 for $K \leq 1200$), the exact enclosures of $a(1), a(2), a(3)$, the closed forms $a(1) = 3 - 4 \ln 2$, etc., $0 < \alpha^* \leq \frac{2}{K+1}$ for $2 \leq K \leq 150$, $K+1 \leq M \leq K+150$, and agreement of the amplitudes of Table 1 with an independent implementation in [1] (all intervals with $n \leq 40$).
- `check_constructions.py`: the renewal squares agree with the block description ($n \leq 24$); exact verification of Definition 4.1 for all intervals of all boxes $n \leq 22$ with the constructions of Table 1 (α^* rounded to a rational with denominator $\leq 10^{30}$); agreement of $\mathbf{T}$ with an independent implementation of the run recursion ($n \leq 30$); the cell conditions (CELL1), (CELL2) with the exact α^* at 50 digits for all 17296 intervals with $1 \leq \ell \leq h \leq n-2$, $n \leq 48$; the closed forms of Section 8; $Y^0 = \mathcal{R}_n^h - \mathcal{R}_n^{\ell-1}$ for $n \leq 20$; and floating-point checks near the top ($K \leq 6$, $n \leq 400$) and for random intervals with n up to 2000.
- `check_thg_generic.py`: Lemmas 5.2–5.5 with random rational $\mathbf{T}, \mathbf{H}, \mathbf{G}$.
- `check_wk.py`: Lemmas 3.9–3.12 and Proposition 4.3 in exact arithmetic for random rational spectra (with ties) with $d \leq 8$ and every k ; Lemma 3.7 and Corollary 3.8 for 300 random thermal instances.
- `check_end_to_end.py`: the full construction of Theorem 1, including the reduction for non-diagonal H , with explicit Givens rotations, for 300 random instances with $d \leq 6$.

- `check_counterexample.py`: the inequalities in the proof of Proposition 11.1 for $4 \leq d \leq 400$ (exact), a numerical test of the interlacing step, and the $d = 3$ example.

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